

Tuesday

UNIT-1 (Imp)

classmate

Date 30/1/26

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INTRODUCTION TO DATA WAREHOUSE

Introduction

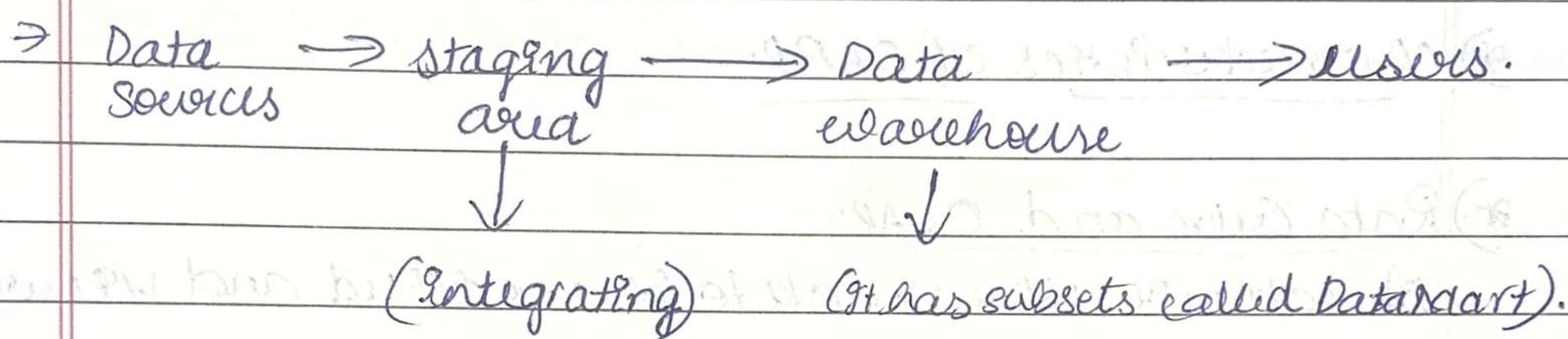
* Major difference between Database (DB) and Data Warehouse (DW)

→ DB is small and DW is very huge.

→ DB query is simple and DW query is complex.

Data: Raw facts.

Warehouse: godown.



→ In staging area we use tool ETL (Extract Transform Load)

→ IBM cloud, Azure, Teradata etc. → warehouse software.
cloud data warehouse.

UNIT-1 — Data Warehousing and Data Mining.

* A data warehouse is a ^(collection) repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site.

Imp * Key Features:-

(and) 4 key features.

1) Subject oriented (particular subject related or stored data) ^{eg: sales}

2) Integrated (combining ^{from different sources} database)

3) Time variant (It has historical data also)

4) Non-volatile (permanent cannot modify).
↓
periodically it will refresh.

* Data Warehousing is the process of constructing and using warehouses.

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Differences between Database and Data Warehouse

(for each side)
 database OLTP -> Online Transaction Processing (database)
 Data Warehouse OLAP -> Online Analytical Processing (data warehouse)
 and * (all) comparison of OLTP and OLAP systems.

Characteristics of OLAP

1) Data Rule and OLAP:

2) Database stores data to be modelled and viewed in multiple dimensions.

• dimension

• fact

3) Dimensions are the entities with respect to which an organisation wants to keep records.

4) Facts are numeric measures.

• each side of details sold

5) Cubes

→ 2D (x,y) axis

→ 3D (x,y,z) axis -> series of 2D

→ 4D -> series of 3D (w,x,y,z) - lattice of cuboid.

Lattice of cuboid

→ general lattice of cuboid for 4-D database (time, item, location, supplier)

→

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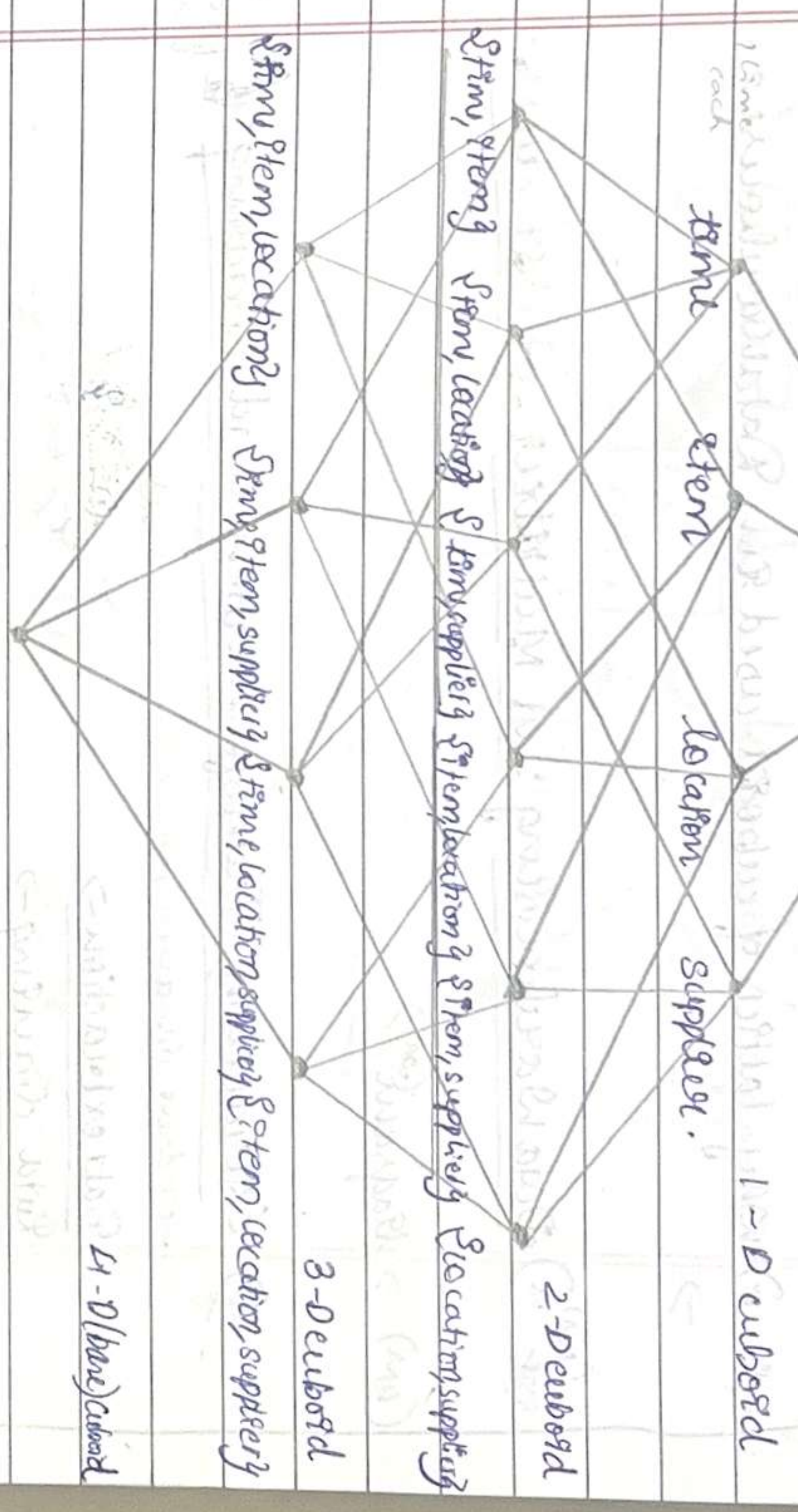
→ cuboid

1 dimension per cuboid -> only 1

$2^n \rightarrow 2^4 > 16$

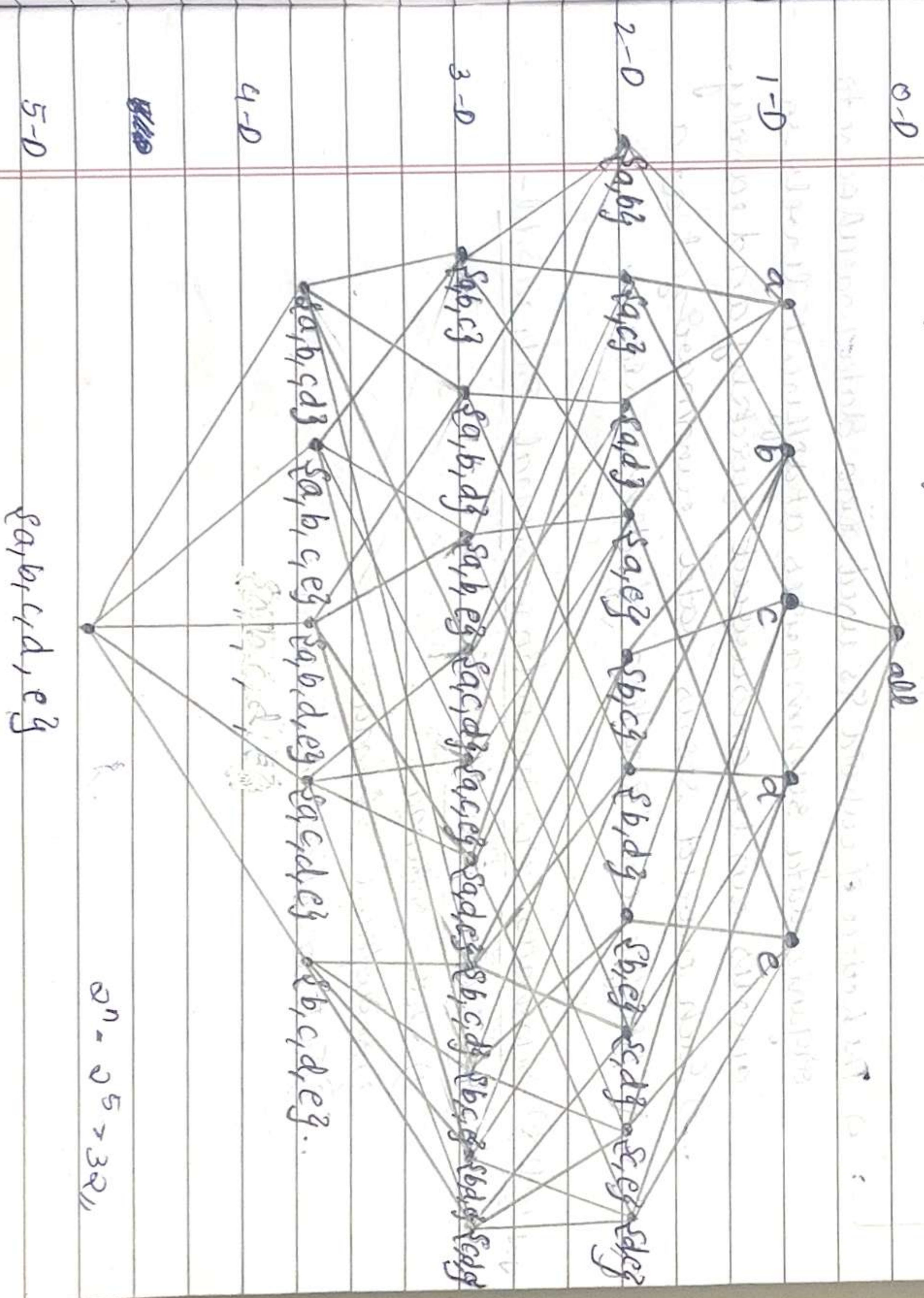
$n \rightarrow$ no. of cuboid (dots)

all -> total sales
 O-D (apex) cuboid



{time, item, location, supplier}

→ lattice of cuboid for dimensions a, b, c, d, e



$2^n = 2^5 > 32$

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No. of cubes per lattice - 2^n .
 $n \rightarrow$ no. of cube.

Q) Why lattice of cube used in Data Warehouse?

(Ans) Data Warehousing: A Multidimensional Architecture -

(Ans) $\rightarrow$ Diagram (Ans)

Q) Extraction, Transformation and Loading (ETL) -

Data extraction $\rightarrow$

Data transformation $\rightarrow$ convert from database (external source) to data warehouse.

Q) The lattice of cube is used in data warehouse to organize data summaries at different levels so queries can be answered quickly and easily.

$\rightarrow$ Each cube stores data summarized at a specific combination of dimensions.

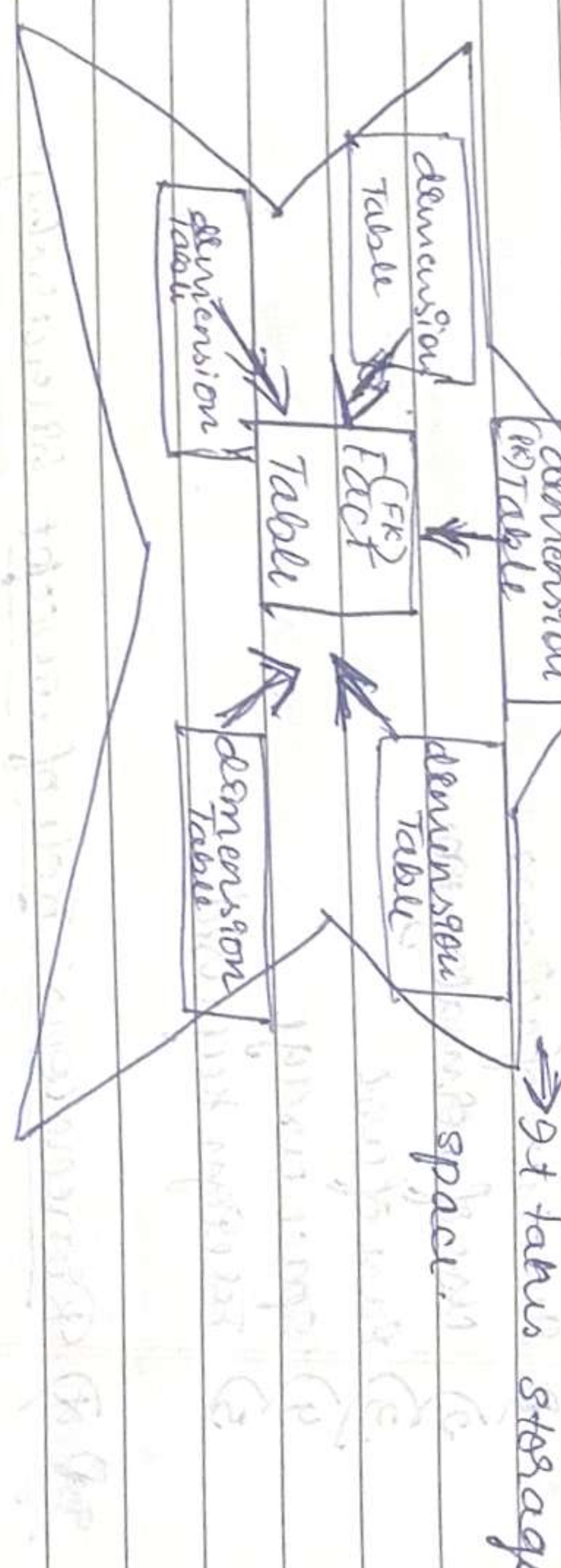
Q) Schemas for Multidimensional Data Model -

- 1) Star Schema
- 2) Snowflake Schema
- 3) Fact constellation schema.

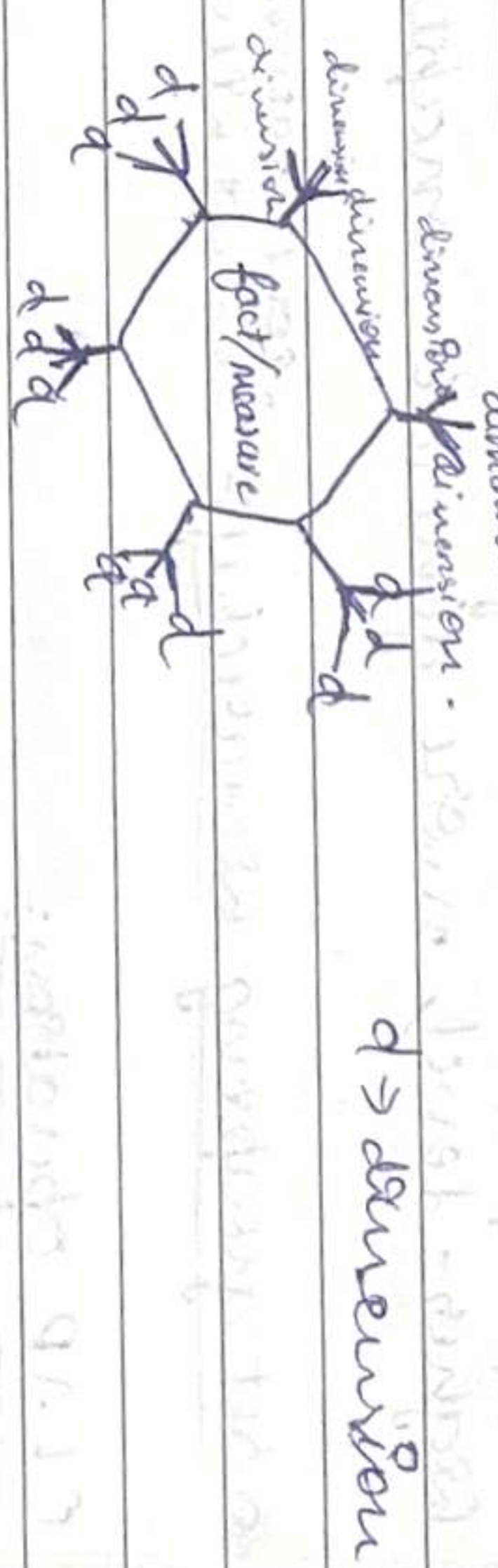
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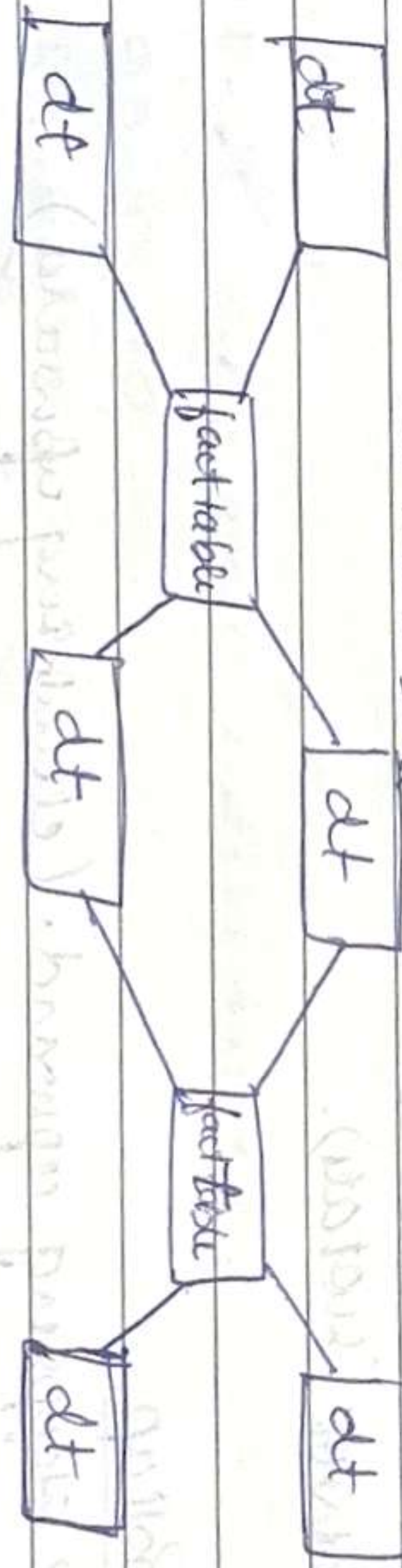
1) Star schema $\rightarrow$ It has ³ redundant data. (redundant data).



2) Snowflake schema - to overcome drawbacks of star schema.



3) Fact constellation schema - (Galaxy Schema).



Q1) Differences between star and snowflake.

- 1) Structure of schema
- 2) Use of normalisation
- 3) Ease of use
- 4) Space usage
- 5) Foreign key used.

Q2) Dimensional: Roll of concept hierarchy:- mapping from low level to high level.

→ A concept hierarchy defines a sequence of mappings from a set of low-level concepts to higher-level, more general concepts.

→ Set grouping hierarchy:- continuation of concept hierarchy

OLAP operations:-

- 1) Roll-up. (deall-up)
- 2) Drill-down
- 3) Slice and dice
- 4) Pivot (rotate).

① Rollup

→ stepping upward. (eliminating upward)

② Drill down

→ stepping downward.

③ Slice and dice.

The slice operation performs a selection on one dimension of the given cube, resulting in a subcube. The dice operation defines a sub cube by performing a selection on two or more dimensions.

Q1) Pivot (Rotate):-

→ Visualization operation that rotates the data exists axes in order to present an alternative data presentation.

• Measure: There are categorization and computation.

→ A data cube measure is a numeric function that can be evaluated at each point in the data cube space.

→ 3 categories based on the aggregate functions used.

- Disaggregation: → sum, min, max, count.
- Aggregation: → avg, sum, count, etc.
- Aggregate: → median, mode, standard deviation.

→ Median: needs full stored data.

10, 18, 25, 40, 55.
Median = 25

② Mode: Needs frequency of all values.

A, B, A, C, A, B.
Mode = A.

③ Rank: depends on entire dataset.

90, 85, 70, 60.
Rank of 70 = 3

Q2) Data Warehouse Models:- (different types).

→ 3 types.

- 1) Enterprise Warehouse (all departments data is stored)
- 2) Data Mart. (subset of data warehouse)
- 3) Virtual Warehouse (virtual subject).

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1) Enterprise Warehouse.

2) Data Mart

3 types

1) Independent data mart.

2) Dependent data mart

3) Virtual Warehouse.

→ Virtual Warehouse is a set of views over operational databases.

* Data Warehouse Design Approaches -

1) Top-down (Datawarehouse → Data Mart) (expensive)

2) Bottom-up (Data Mart to Data Warehouse) (cheap)

3) Incremental and evolutionary approach

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UNIT-II - CHAPTER-1

DATA MINING

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1) Data Mining is a process of discovering interesting patterns and knowledge from large amount of data.

→ KDD (Knowledge Discovery from data) → subprogram of data Mining.

→ Input → Data → Data Mining → Post Processing → Information.

2) Data Mining Tasks:

→ 2 major categories.

1) Predictive tasks. → predicting something (future of some attributes)

2) Descriptive tasks → coming up with pattern.

→ Predictive tasks:- (classification)

→ The objective of this task is to predict the values of a particular attribute based on the value of other attributes.

2) Descriptive tasks:- (use for processing)

→ The objective is to discover patterns that summarise the relationship in the data.

* Four core data mining tasks:-

1) Predictive Modeling → Classification

2) Association analysis

3) Clustering

4) Outlier

1) Predictive Modelling - (predictive tasks)

→ predicting discrete variable.
 (classified) a) classification - If 25 linked for discrete target variable
 (ordinal) b) Regression - If 25 linked for continuous target variable
 (continuous)

→ predicting continuous variable

eg:- Predicting the type of a person's

→ rough diagram (use should draw) graph.

2) Association analysis - (descriptive tasks)

→ ideally → market based analysis.

→ In goal of association analysis is to extract the most of interesting patterns in an efficient manner.

→ Market basket transaction
 (at least 5 we should use)

3) Cluster analysis - (descriptive tasks)

→ used to identify data objects that are similar to one another.

eg:- news papers article.

4) Anomaly Detection - (Descriptive tasks)

→ It is the task of identifying observations whose characteristics are significantly different from the rest of the data. Such observations are known as anomalies or outliers.

→ Fraud detection, network intrusion,

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* Types of Data -

1) What is Data Set?

→ Selection of data objects and their attributes.

→ An attribute is a property or characteristic of an object.

* Data can be classified into two broad types:

1) Quantitative - eg:- age, size

2) Qualitative - eg:- color, gender, green, yellow, blue

* Types of Attribute:

1) Nominal - Qualitative

2) Ordinal

3) Interval - Quantitative

4) Ratio

1) Nominal - (name)

→ Nominal means "relating to names".

eg:- Gender and marital status.

2) Ordinal - (order)

→ An ordinal attribute is an attribute with possible values that have a meaningful order or ranks.

eg:- grade (A+, A, A-, B+, and so on).

3) Interval - (difference between values)

→ Interval - scaled attributes are measured on a scale of equal-size units.

→ attribute allows us to compare and quantify the difference between values.

eg:- difference between 2:00pm and 3:00pm (1h).

1) Ratio scaled attributes: (multiple $\rightarrow$ fraction)

$\rightarrow$ A measurement is ratio-scaled, we can speak of value as being a multiple (or ratio) of another value.
eg: ₹60,000 is twice ₹30,000.

2%)

* Assessing attributes by the number of values

1) Discrete attributes (eg: marital status).
 $\rightarrow$ has only finite set of values.

2) Binary attributes (eg: exam result).
 $\rightarrow$ assume only 2 values.

3) Continuous attributes (eg: employee income).
 $\rightarrow$ has a real numbers as attribute values.

* Data Quality: (all terminology for IM).
1) Noise and artifacts:

$\rightarrow$ Noise is a random error or variance in a measured variable.

$\rightarrow$ Distortion of the data are referred as artifacts (instead of noise but by an overall because of noise that is called artifacts).

2) Outliers:

$\rightarrow$ Outliers are data objects that have characteristics that are different from most of the other data objects in the data set.

$\rightarrow$ noise which is not needed, outliers may be sometimes in use.

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3) Inconsistent values:

$\rightarrow$ It is wrong data. (e.g. age, height, weight, etc.)
eg: $\rightarrow$ age cannot be negative.

4) Missing values:

$\rightarrow$ Information is not collected.

$\rightarrow$ Here to fill missing values $\rightarrow$ (mean/median/mode/variable)
• ignore the tuple or attribute.
• fill the missing value manually.
• Use a global constant to fill in the missing value. ("unknown" or "null")
• Use a measure of central tendency for the attribute. (mean or median).
• Use the most probable value to fill in the missing value. (decision tree).

5) Duplicate data:

$\rightarrow$ A data set may include data objects $\rightarrow$ duplication.

6) Precedence, Bias and Accuracy:

$\rightarrow$ Precedence: The closeness of repeated measurements to one another.
 $\rightarrow$ Bias: Measure of how far the expected value is from the true value.
 $\rightarrow$ Accuracy: closeness of the measurement to the true value.

*) Types of data sets:

- 1) Record data
- 2) Graph based data
- 3) Ordered data

9) Related data -

(one source of transaction)

• Transaction data (Market basket data)

• Data Matrix (Pattern matrix)

• Document term matrix (Sparse data matrix)

(Matrix no. of zeroes)

9ii) Related data

• Sequential data

• Temporal data

• Spatial data

*) Graph based data

→ Data represented using graph

→ Data with relationship among effects

→ Data with effects that are graphs

*) Two types of graph based data

• Ordered data

→ Data sets are in same order

→ Sequential data (Temporal data)

→ Time is associated with the data

• Time series data

→ Data sets are collected based on time interval

• Spatial data

→ based on location

*) Characteristics of a data sets -

→ new characteristics that apply to many data sets and have a significant impact on the data mining technique

Characteristics of a data sets -

① Sparsity

② Resolution

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① Dimensionality - columns, attributes, variables.

② types:-

• High dimensionality - complex analysis.

• Low dimensionality

② Sparsity - more number of 0's when there are empty data.

③ Resolution - level of data (how detailed your data is).

↓

High resolution

(detailed data)

Low resolution

(not so detailed data)

↓

*) KDD - Knowledge Discovery from Data.

It is the process of extracting data.

② Diagram - (both diagram we should draw & explain).

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*) Important spreads for OP are:

- 7) Agglutination :-

2) Damp blange

- 1) Simple random sample without replacement (SRSW)
- 2) Simple random sample with replacement (SRSWR)
- 3) Cluster sample with replacement (SRSCW)
- 4) Stratified sample (Random selection from each cluster)

(*) Sample sizes - if n is good if size is large.
Progressive sampling - start with small sample size and then gradually increase it from each obsn.

→ At 2D a process of summarizing the information in a dataset while preserving important information (3D → 2D → 1D).

2 types:

d) loaders. If we can construct a dimer^{e} reduction

2) easy. if we can't construct after preimage

(147) * number of dimensionality

→ The curse of dimensionality applies to the problems that arise when the number of dimensions (features) in a dataset increases, making analysis difficult.

c) Feature subset selection

→ Feature means attribute.

It $\Rightarrow$ Feature subset selection is the process of selecting a subset of relevant features from the original dataset and removing redundant or irrelevant features.

$$\rightarrow \underline{\underline{27 \text{ Pa}}}$$

9) Redundant Feature. (duplicate data)

ii) Irrelevant Feature (not useful data)

→ Three standard approaches are:

- blackbox ^{approaches}
 - disappearing approach // seen based on no. of sets
 - Fuzzy approach // seen only once
 - Embedded approach // seen based on user satisfaction of both

- *Crappus claus*

$$S = \{a, b, c\}$$

$\{ \{a, b\}, \{c\}, \{a, b\}, \{a, c\} \}$

5abc3

* Feature Mapping

→ Alternative to eliminate features.

1) Invertible Transformation:

→ An attribute transform as a function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values.

→ Two types:

- 1) Simple functional transformation.
- 2) Normalization and standardization

a) Simple Functions:

if x is a variable,

$$e^x$$

$$x^k$$

$$\log x$$

$$1/x$$

$$\sqrt{x}$$

$$\sin x$$

$$\cos x$$

$$\tan x$$

$$\cot x$$

$$\sec x$$

$$\csc x$$

→ min-max normalization

$$V' = \frac{V - \min_A}{\max_A - \min_A} \quad (\text{new max} - \text{new min}) + \text{new min}$$

→ z-score normalization

$$V' = \frac{V - \bar{A}}{\sigma_A}$$

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* Min-max normalization:

① Suppose that the minimum and maximum values for the attribute income are \$12,000 and \$98,000 respectively.

• We would like to map income to the range [0.0, 1.0].

By min-max normalization, a value of \$73,000 for income is transformed to

$$= \frac{73000 - 12000}{98000 - 12000} (1.0 - 0) + 0 = 0.716$$

* z-score normalization (zero-mean normalization):

① Suppose that the mean and standard deviation of the values for the attribute income are \$24,000 and \$16,000, respectively. With z-score normalization, a value of \$73,000 for income is transformed to

$$= \frac{73000 - 24000}{16000} = 1.8125$$

* Measures of similarity and dissimilarity:

→ Similarity is used to refer to either good similarity or dissimilarity between the objects

→ Dissimilarity - between two objects is a numerical measure of the degree to which the two objects are alike.

→ (no similarity) $1 \rightarrow$ (complete similarity).

$|x-y| = \text{absolute value}$

(distance)
→ Distance is a numerical measure of the degree to which the two objects are different.

* Dissimilarities between data objects -

→ Euclidean distance -

Euclidean distance, d , between two points X and Y in one-, two-, three-, or higher dimensional space is -

* Minkowski distance -

$$d(x, y) = \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r}$$

$r=1$ → Manhattan / L1 norm

$r=2$ → Euclidean distance ($L_2 \rightarrow \sqrt{\quad}$) / L2 norm

$r=\infty$ → Chebyshev / L ∞ norm

→ Euclidean distance -

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

distance b/w $P1 \in P2$.

• $P1(0, 2)$ • $P2(2, 0)$

$$\sqrt{(0-2)^2 + (2-0)^2}$$

$$\sqrt{(-2)^2 + (2)^2}$$

$$= \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} = 2.828$$

Distance is always positive.

• $P1(0, 2)$ • $P2(3, 1)$

$$\sqrt{(0-3)^2 + (2-1)^2}$$

$$= \sqrt{(-3)^2 + (1)^2}$$

• $P1(0, 2)$ • $P2(3, 1)$

$$= \sqrt{9+1} = \sqrt{10}$$

* Properties of Euclidean distance -

1)

Semimetric measure for binary data

2) Methods

- 1) SMC (Simple Matching coefficient)
- 2) Jaccard

① SMC $\rightarrow$ frequency

SMC = No. of matching attribute values

$$\frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}}$$

② Jaccard

J = no. of matching presence / no. of attributes not involved in 0 or 1 matches

$$\frac{f_{11}}{f_{01} + f_{10} + f_{11}}$$

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Let us find the no. of attributes in X_1, X_2, X_3, X_4, X_5 .
for the no. of attributes in X_1, X_2, X_3, X_4, X_5 .
for the no. of attributes in X_1, X_2, X_3, X_4, X_5 .

Let us find the no. of attributes in X_1, X_2, X_3, X_4, X_5 .

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$$\begin{array}{r} 42 \times 114 \\ \underline{42 \times 114} \\ 4788 \end{array}$$

$$\begin{array}{r} 25 \\ \underline{25} \\ 15.81 \end{array}$$

$$\cos(x, y) = 0.32$$

5) Feature Functions

- New Methodology:
 - 1) Feature extraction
 - 2) Mapping data to a new space.
 - 3) Feature construction.

① Feature extraction

→ creation of new set of features from the original raw data is known as feature extraction.
eg: range → auto pixels.

5				
Pixel 1	Pixel 2	...	Pixel 784	
0	255	120	...	60

- ② Mapping data to a new space -
→ Sometimes a totally different view of data can provide important and interesting data.

Student	marks
Alice	92

Mark range	grade
90-100	A

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③ Feature Construction

Student	Math	Science	Eng.
Alice	80	75	85

Student	dogmates (Feature)
Alice	80

→ forming new column ~~from~~

④ Discretization and Binarization

→ Discretization is the process of converting a continuous attribute into an ordinal (categorical) attribute.

→ A potentially infinite no. of values are mapped into a small no. of categories.

eg: Height measured as Short, medium, large?
Age measured as Youth, middle-aged, senior?

→ Binarization maps a continuous or

categorical (ordinal) attribute into one or more attributes.

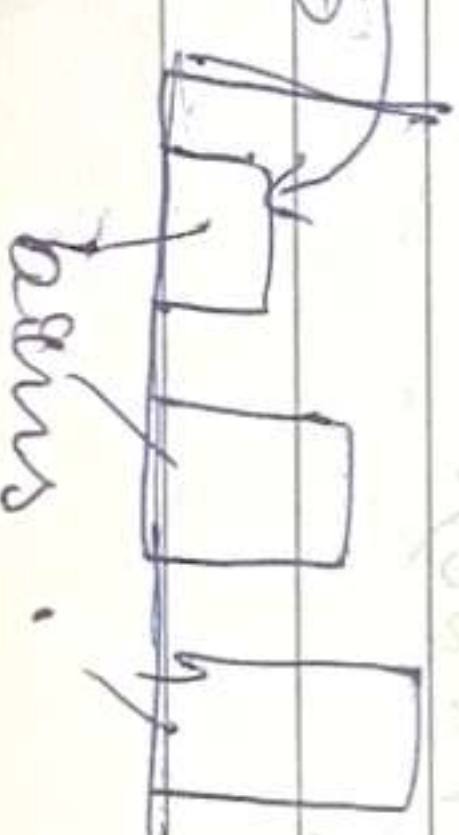
→ often convert continuous attribute into categorical then categorical to binary.

→ Two types of Discretization -

- 1) Supervised (with class label).
- 2) Unsupervised (without class label).

→ Equal-width
→ Equal-Frequency.

→ bins →





* Equal width interval description

$$\text{Interval} = \frac{V_{\max} - V_{\min}}{k}$$

$$\text{Boundaries} = V_{\min} + (k \times \text{interval})$$

equal width	equal continuous values	sorted values
10	50 15 20 12 25 40 30	10 12 15 20 25 30 40 50

$$\text{Interval} = \frac{50 - 10}{2} = 20$$

$$k = \frac{40}{2} = 20$$

$$\text{Boundaries} = 10 + (1 \times 20)$$

$$= 30$$

$$k = 2$$

discrete values	1	2	1	1	1	1	2	1
-----------------	---	---	---	---	---	---	---	---

equal frequency description

So interval will be $8/2 = 4$ data values.

discrete value	1	2	1	1	1	2	2	2
----------------	---	---	---	---	---	---	---	---

Measuring challenges of data measuring:-

- 1) Scalability.
- 2) High dimensionality.
- 3) Heterogeneous and complex data.
- 4) Data ownership and distribution.
- 5) Non-traditional analysis. (1st experience friend)

UNIT-3

ASSOCIATION ANALYSIS.

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Association Analysis:-

- It is useful for discovering interesting relationships in large dataset.
- Relationship can be represented in the form of association rules and
- discovery $\rightarrow$ $\{B \rightarrow C\}$
- Binary Representation:-
- Market basket data can be represented in a binary format.

- Itemset:-
- An association analysis a collection of zero or more items treated as itemset.
- Transaction with its items defined as the number of items present in a transaction.

Association rule:-

$X \rightarrow Y$. disjoint $\rightarrow X \cup Y \rightarrow [X \cup Y = \emptyset]$
 $X \rightarrow Y$ is an item set
 $Y \rightarrow$ an item set.

Support and Confidence.

- Support, $S(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{N}$. $\sigma \rightarrow$ support count
 $N \rightarrow$ no. of transactions
- Confidence, $C(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)}$.

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Example for following rules

Example for following rules $\rightarrow$ $\{B \rightarrow C\}$.
 Items, diapers, beer $\rightarrow 2$.
 Items, diapers, beer $\rightarrow 5$.
 total no. of transaction $= 5$.

Support, $S(X \rightarrow Y)$.
 $\frac{2}{5} = 0.4$
 40%
 Confidence, $C(X \rightarrow Y)$
 $\frac{2}{3} = 0.67$
 67%

Support $= \frac{\sigma(X \cup Y)}{N}$.
 $= \frac{\sigma(M, D, B)}{5} = \frac{2}{5} = 0.4 = 40\%$

Confidence $= \frac{\sigma(X \cup Y)}{\sigma(X)}$.
 $= \frac{\sigma(M, D, B)}{\sigma(M, D)} = \frac{2}{3} = 0.67 \rightarrow 67\%$

Example

Support $= A \rightarrow C$
 $\frac{2}{5} = 0.4 = 40\%$

Confidence $= \frac{\sigma(A \rightarrow C)}{\sigma(A)}$
 $\frac{2}{3} = 66.6\%$

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C → A.

→ $\frac{\text{Support}_2}{5} = 50\%$

→ confidence = $\frac{8}{8} = 100\%$

* Association rule discovery:

→ Given a set of transactions, the goal of association rule mining is to find all rules having support ≥ min sup threshold.

• confidence ≥ min conf threshold.

* Brute force approach for mining association rules:

→ generating all possible rule using

$R = 3^d - 2^{d+1} + 1$

$R = 3^6 - 2^7 + 1$

→ 600 rules.

→ Brute force approach is not recommended

(Time consuming)

* Association rule mining algorithm decompose

the problem into two major approaches.

• Frequency itemset generation → generating frequent itemsets

• Rule generation → extracting strong rule.

→ An ordered lattice structure is called candidate itemset.

→ Support should be greater than or equal to minsup

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* Apriori Principle:

→ If an itemset is frequent, then all of its subsets must also be frequent.
→ If an itemset is infrequent, then all of its supersets must also be infrequent.
→ Generally frequent itemsets for given transactional data

are, minsup = 2 using Apriori Principle.

Itemset	Support	Is frequent?
I1	5	Yes
I2	4	Yes
I3	3	Yes
I4	2	Yes
I5	2	Yes
I1, I2	3	Yes
I1, I3	2	Yes
I1, I4	2	Yes
I1, I5	2	Yes
I2, I3	2	Yes
I2, I4	2	Yes
I2, I5	2	Yes
I3, I4	2	Yes
I3, I5	2	Yes
I4, I5	2	Yes
I1, I2, I3	2	Yes
I1, I2, I4	2	Yes
I1, I2, I5	2	Yes
I1, I3, I4	2	Yes
I1, I3, I5	2	Yes
I1, I4, I5	2	Yes
I2, I3, I4	2	Yes
I2, I3, I5	2	Yes
I2, I4, I5	2	Yes
I3, I4, I5	2	Yes
I1, I2, I3, I4	2	Yes
I1, I2, I3, I5	2	Yes
I1, I2, I4, I5	2	Yes
I1, I3, I4, I5	2	Yes
I2, I3, I4, I5	2	Yes
I1, I2, I3, I4, I5	2	Yes

Candidate 1 itemset

Frequent 1 itemset

Itemset	Support	Is frequent?	Candidate 2 itemset	Support	Is frequent?
I1, I2	3	Yes	I1, I2, I3	2	Yes
I1, I3	2	Yes	I1, I2, I4	2	Yes
I1, I4	2	Yes	I1, I2, I5	2	Yes
I1, I5	2	Yes	I1, I3, I4	2	Yes
I2, I3	2	Yes	I1, I3, I5	2	Yes
I2, I4	2	Yes	I1, I4, I5	2	Yes
I2, I5	2	Yes	I2, I3, I4	2	Yes
I3, I4	2	Yes	I2, I3, I5	2	Yes
I3, I5	2	Yes	I2, I4, I5	2	Yes
I4, I5	2	Yes	I3, I4, I5	2	Yes
I1, I2, I3	2	Yes	I1, I2, I3, I4	2	Yes
I1, I2, I4	2	Yes	I1, I2, I3, I5	2	Yes
I1, I2, I5	2	Yes	I1, I2, I4, I5	2	Yes
I1, I3, I4	2	Yes	I1, I3, I4, I5	2	Yes
I1, I3, I5	2	Yes	I1, I4, I5	2	Yes
I1, I4, I5	2	Yes	I2, I3, I4	2	Yes
I1, I5, I4	2	Yes	I2, I3, I5	2	Yes
I2, I3, I4	2	Yes	I2, I3, I5	2	Yes
I2, I4, I5	2	Yes	I2, I5, I4	2	Yes
I3, I4, I5	2	Yes	I3, I5, I4	2	Yes
I1, I2, I3, I4	2	Yes	I1, I2, I3, I5	2	Yes
I1, I2, I4, I5	2	Yes	I1, I2, I5, I4	2	Yes
I1, I3, I4, I5	2	Yes	I1, I4, I5	2	Yes
I2, I3, I4, I5	2	Yes	I2, I4, I5	2	Yes
I3, I4, I5	2	Yes	I3, I5, I4	2	Yes
I1, I2, I3, I4, I5	2	Yes	I1, I2, I3, I4, I5	2	Yes

Steps:

① → Create candidate 1 itemset & find support.

② → Prune itemset by comparing with minsup.

③ → Create candidate 2 itemset (combination) & support.

④ → Prune concept.

Tuesday

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Q1) 4 items I1, I2, I3, I4
Item 1 has support count
Item 2 has support count

Item 1 has support count 50%, Item 2 has support count 60%

Item 3 has support count 70%, Item 4 has support count 80%

Item 5 has support count 90%, Item 6 has support count 100%

Item 7 has support count 110%, Item 8 has support count 120%

Item 9 has support count 130%, Item 10 has support count 140%

Item 11 has support count 150%, Item 12 has support count 160%

Item 13 has support count 170%, Item 14 has support count 180%

Item 15 has support count 190%, Item 16 has support count 200%

Item 17 has support count 210%, Item 18 has support count 220%

Item 19 has support count 230%, Item 20 has support count 240%

Item 21 has support count 250%, Item 22 has support count 260%

Item 23 has support count 270%, Item 24 has support count 280%

Item 25 has support count 290%, Item 26 has support count 300%

candidate 2 itemset

itemset support count

I1, I2 4

I1, I3 4

I1, I4 1

I1, I5 2

I2, I3 4

I2, I4 2

I2, I5 2

I3, I4 1

I3, I5 0

I4, I5 0

I1, I2, I3 1

I1, I2, I4 1

I1, I2, I5 1

I1, I3, I4 1

candidate 3 itemset

itemset support count

I1, I2, I3 1

I1, I2, I4 1

I1, I2, I5 1

I1, I3, I4 1

I1, I3, I5 1

I1, I4, I5 1

I2, I3, I4 1

I2, I3, I5 1

I2, I4, I5 1

I3, I4, I5 1

I1, I2, I3, I4 1

I1, I2, I3, I5 1

I1, I3, I4, I5 1

I2, I3, I4, I5 1

candidate 4 itemset

itemset support count

I1, I2, I3, I4 1

I1, I2, I3, I5 1

I1, I2, I4, I5 1

I1, I3, I4, I5 1

I2, I3, I4, I5 1

I1, I2, I3, I4, I5 1

I1, I2, I3, I4, I5 1

I1, I2, I3, I4, I5 1

I1, I2, I3, I4, I5 1

I1, I2, I3, I4, I5 1

I1, I2, I3, I4, I5 1

I1, I2, I3, I4, I5 1

I1, I2, I3, I4, I5 1

I1, I2, I3, I4, I5 1

Support threshold = 50% $\Rightarrow 0.5 \times 6 = 3 \Rightarrow \text{minsup} = 3$.

$$\frac{S \cdot \sigma \cdot X \cdot Y}{N}$$

$$0.5 \times \frac{X \cdot Y}{6}$$

• Candidate 1 itemset:

itemset	support	Frequent 1 itemset
$\{I1\}$	4	$\{I1\}$
$\{I2\}$	5	$\{I2\}$
$\{I3\}$	4	$\{I3\}$
$\{I4\}$	4	$\{I4\}$
$\{I5\}$	2	discarded

became
of less
than
minsup

• Candidate 2 itemset:

itemset	support	Frequent 2 itemset
$\{I1, I2\}$	4	$\{I1, I2\}$
$\{I1, I3\}$	3	$\{I1, I3\}$
$\{I1, I4\}$	2	discarded (because < minsup)
$\{I2, I3\}$	4	$\{I2, I3\}$
$\{I2, I4\}$	3	$\{I2, I4\}$
$\{I3, I4\}$	2	discarded (because < minsup)

• Candidate 3 itemset:

itemset	support	Frequent 3 itemset
$\{I1, I2, I3\}$	3	$\{I1, I2, I3\}$
$\{I2, I3, I4\}$	2	discarded
$\{I1, I2, I4\}$	2	discarded (because < minsup)

because of
apriori principle

• Candidate 4 itemset:

$\Rightarrow$ The frequent itemsets are $\{I1, I2, I3\}$

3

Find frequent itemset for given transactional data $\rightarrow$ support count is 2 and min-conf value is 60%.

TID	items	support
T1	$\{1, 3, 4\}$	3
T2	$\{2, 3, 5\}$	3
T3	$\{1, 2, 3, 5\}$	4
T4	$\{2, 3, 5\}$	3
T5	$\{1, 3, 5\}$	2

$$\frac{S \cdot \sigma \cdot (X \cdot Y)}{N}$$

$\Rightarrow$

Candidate 1 itemset:

itemset	support	Frequent 1 itemset
$\{1\}$	3	$\{1\}$
$\{2\}$	3	$\{2\}$
$\{3\}$	4	$\{3\}$
$\{4\}$	1	discarded < min 2
$\{5\}$	4	$\{5\}$

• Candidate 2 itemset:

Frequent 2 itemset:

itemset	support	Frequent 2 itemset
$\{1, 2\}$	1	discarded (than 2)
$\{1, 3\}$	3	$\{1, 3\}$
$\{1, 5\}$	2	$\{1, 5\}$
$\{2, 3\}$	2	$\{2, 3\}$
$\{2, 5\}$	3	$\{2, 5\}$
$\{3, 5\}$	3	$\{3, 5\}$

candidate's stemset - Frequent stemset

stemset	support	→	stemset	support
$\{1, 3, 5\}$	2		$\{1, 3, 5\}$	2
$\{2, 3, 5\}$	2		$\{2, 3, 5\}$	2

According to support threshold of an itemset is frequent then all its supersets are also frequent.

In the previous step $\{1, 2\}$ is discarded, now in candidate 3 stemset, there is no stemset with $\{1, 2\}$, hence no stemset is discarded.

Conclusion → The frequent itemset are $\{1, 3, 5\}$ and $\{2, 3, 5\}$

Conclusion $(X \neq Y)$ Rule generator

The frequent itemset are

$\{1, 3, 5\}$ and $\{2, 3, 5\}$.

For $\{1, 3, 5\}$, subsets are $\{1, 3\}, \{1, 5\}, \{3, 5\}$.

$\{1, 3\} \rightarrow \{1, 3\}$ confidence = $2/3 = 66.6\%$ (Selected)

$\{1, 5\} \rightarrow \{1, 5\}$ confidence = $2/2 = 100\%$ (Selected)

$\{3, 5\} \rightarrow \{3, 5\}$ confidence = $2/3 = 66.6\%$ (Selected)

$\{1\} \rightarrow \{1, 3, 5\}$ confidence = $2/3 = 66.6\%$ (Selected)

$\{3\} \rightarrow \{1, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

$\{5\} \rightarrow \{1, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

$\{1, 3\} \rightarrow \{1, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

$\{1, 5\} \rightarrow \{1, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

$\{3, 5\} \rightarrow \{1, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

$\{1, 3, 5\} \rightarrow \{1, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

$\{2, 3, 5\} \rightarrow \{2, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

$\{1, 2, 3, 5\} \rightarrow \{1, 2, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

$\{1, 2, 3, 5\} \rightarrow \{1, 2, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

$\{1, 2, 3, 5\} \rightarrow \{1, 2, 3, 5\}$ confidence = $2/4 = 50\%$ (Rejected)

Thursday

major difference (b) is a priori & (1) is a posteriori
→ can't date generation

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* FP Growth algorithm - Frequent Pattern Growth

TID	Items	Ordered-itemset	support - 2/0.1
1	a, b	a, b	1
2	b, c, d	b, c, d	1
3	a, c, d, e	a, c, d, e	1
4	a, d, e	a, d, e	1
5	a, b, c	a, b, c	1
6	a, b, c, d	a, b, c, d	1
7	a	a	1
8	a, b, c	a, b, c	1
9	a, b, d	a, b, d	1
10	b, c, e	b, c, e	1

generate the frequent itemset for given transactional data.
using minsup = 2 using FP growth algorithm.

2 steps in FP growth

1) Construct FP tree

2) Traverse the FP tree to generate frequent itemsets

Item	supportcount
a	8
b	7
c	6
d	5
e	3

→ arrange supportcount with minsup.

→ arrange in descending order first.

→

explain FP algorithm (take exam eg)
• construct FP tree.
• explain options possible.
• find frequent itemset & null generation.

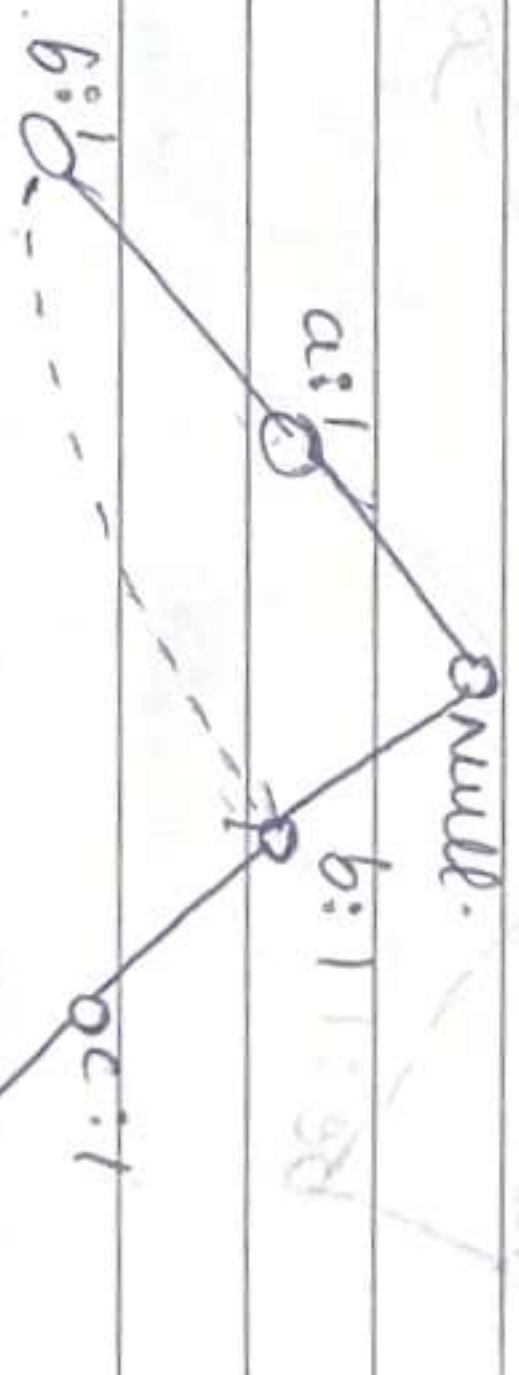
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Item	supportcount
a	8
b	7
c	6
d	5
e	3

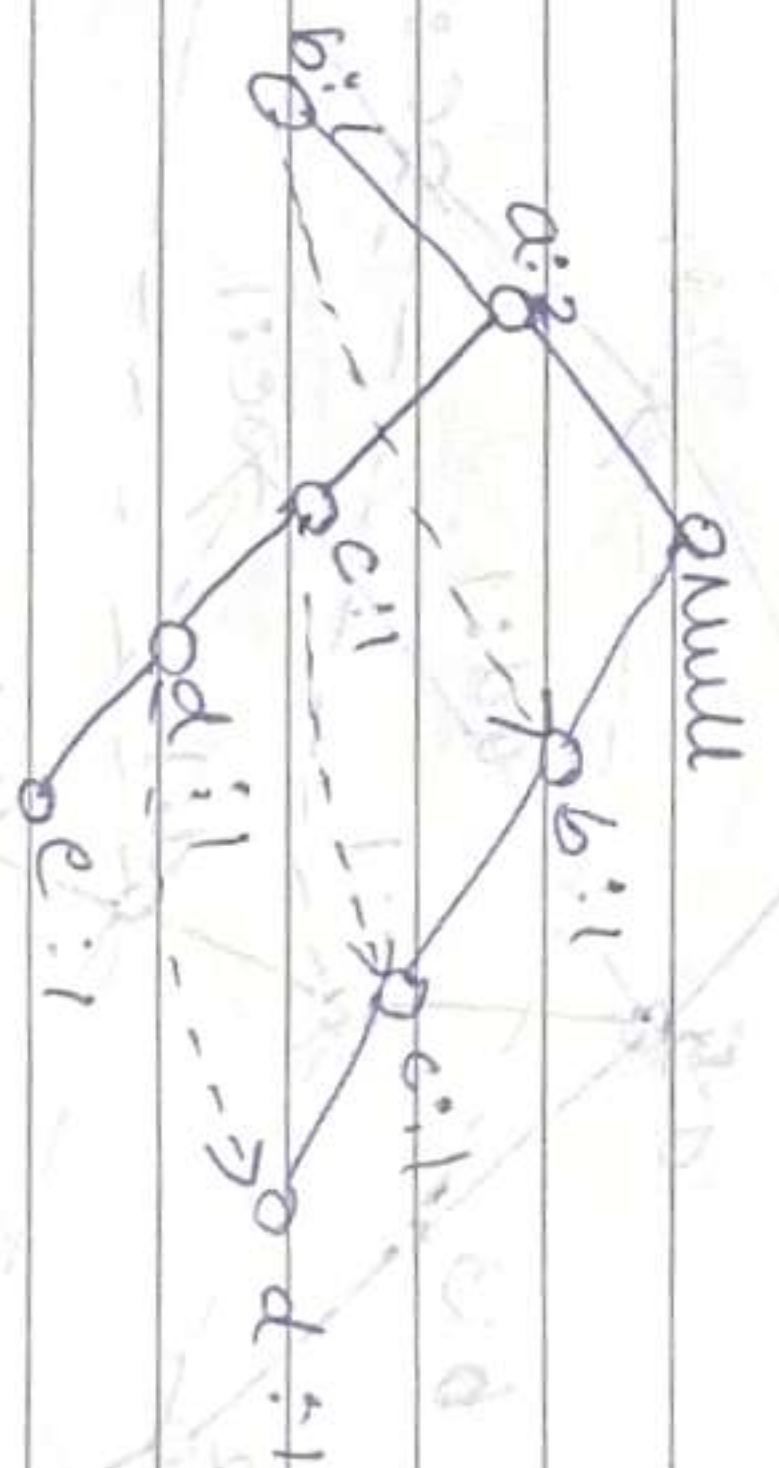
After TID = 1.



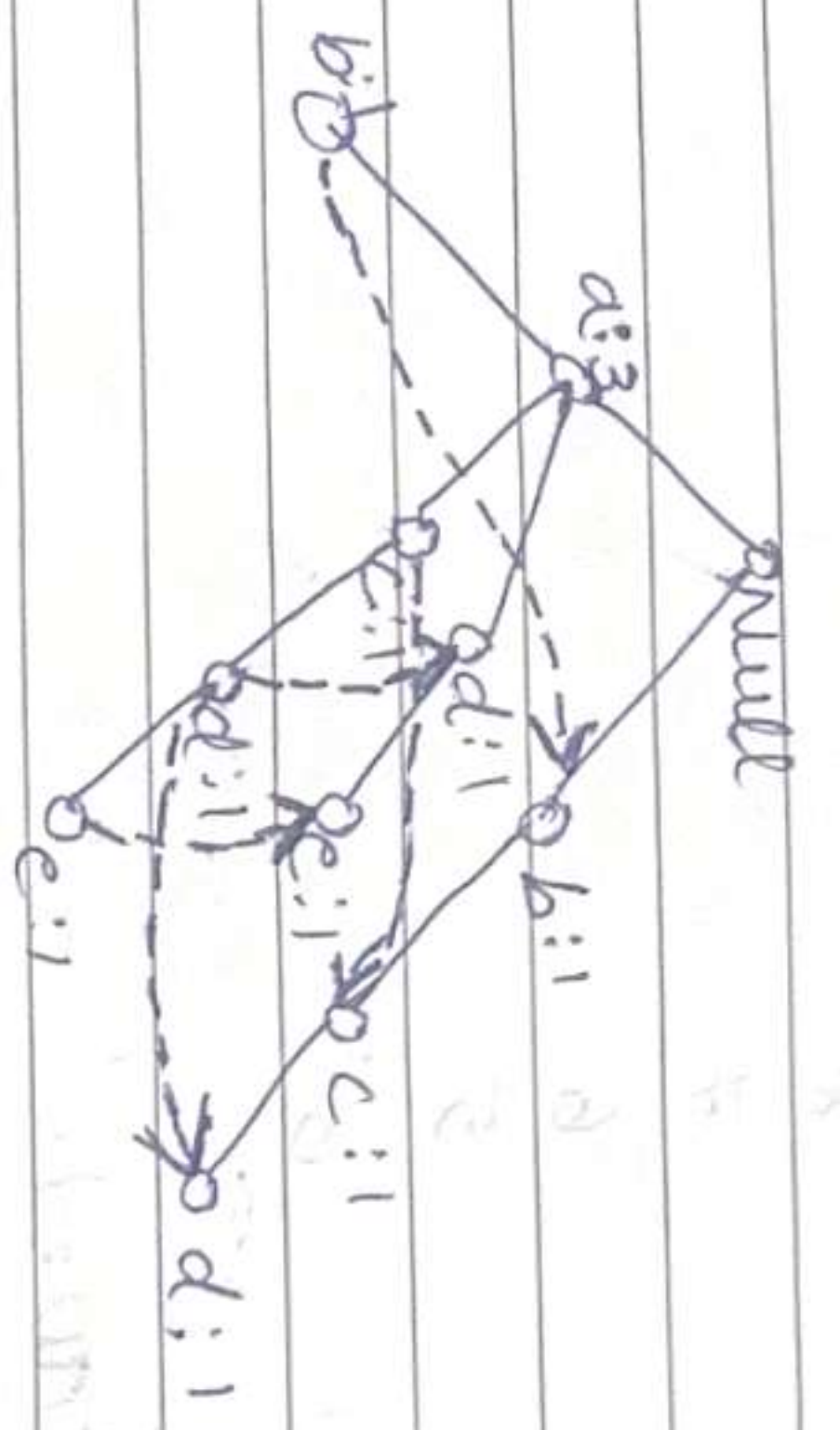
After TID = 2.



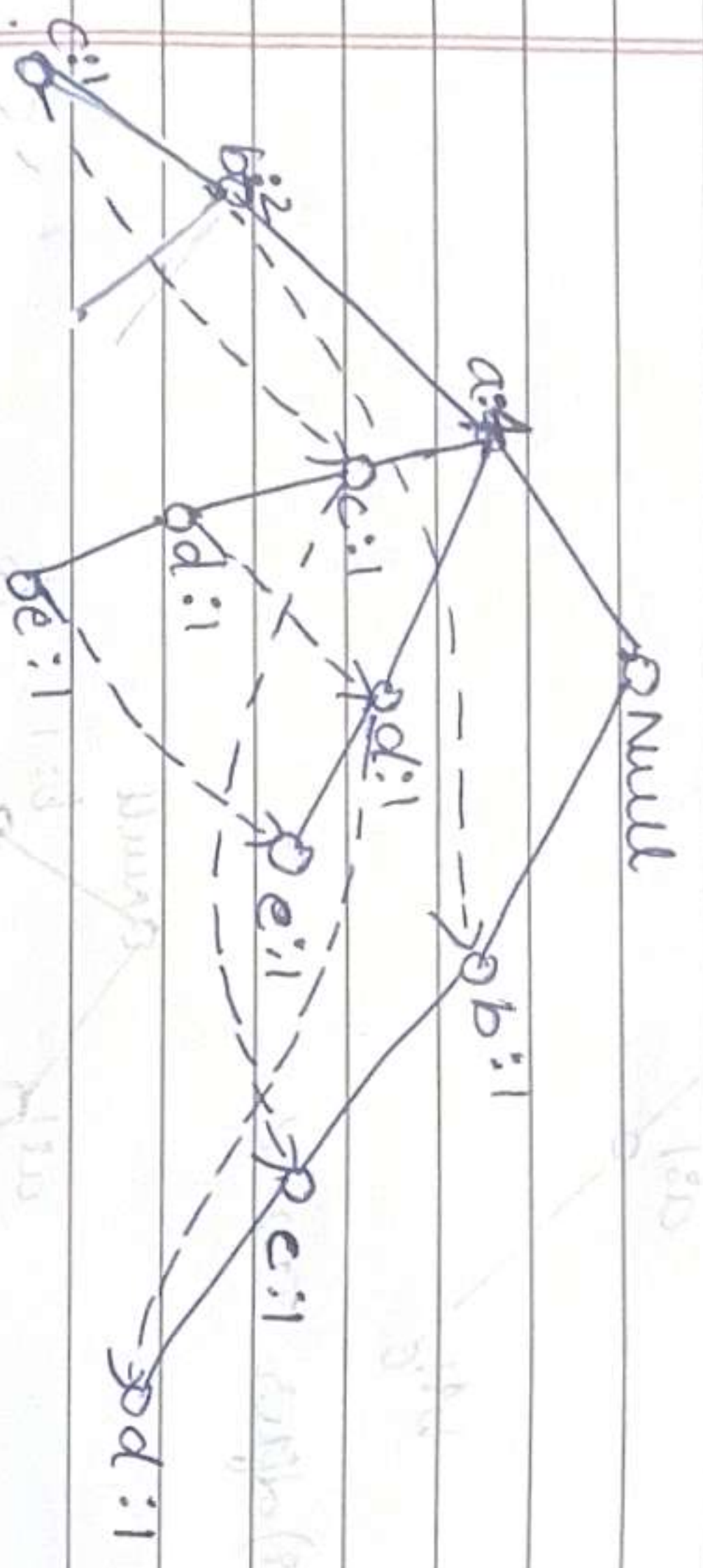
After TID = 3.



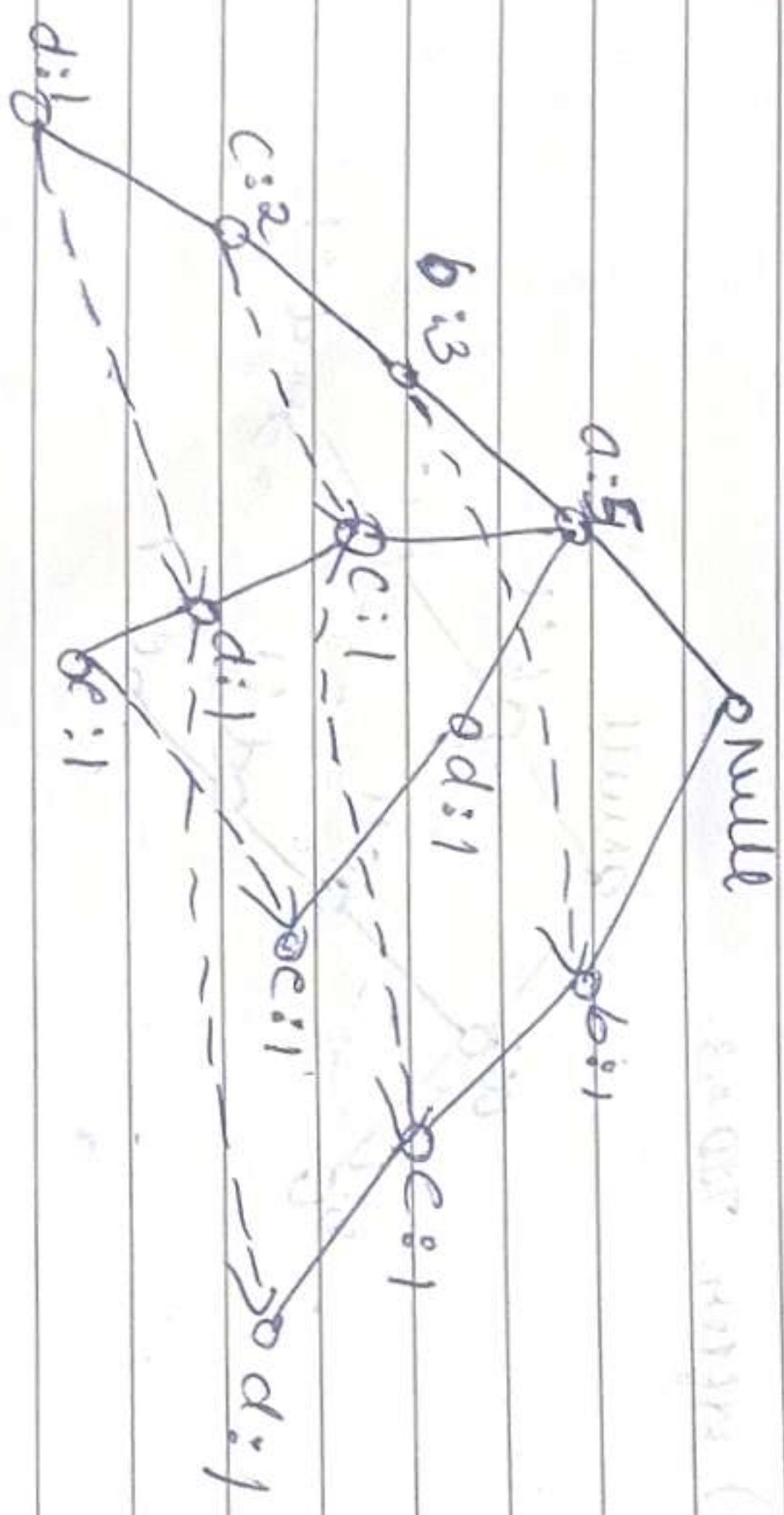
iv) after TID = 4.



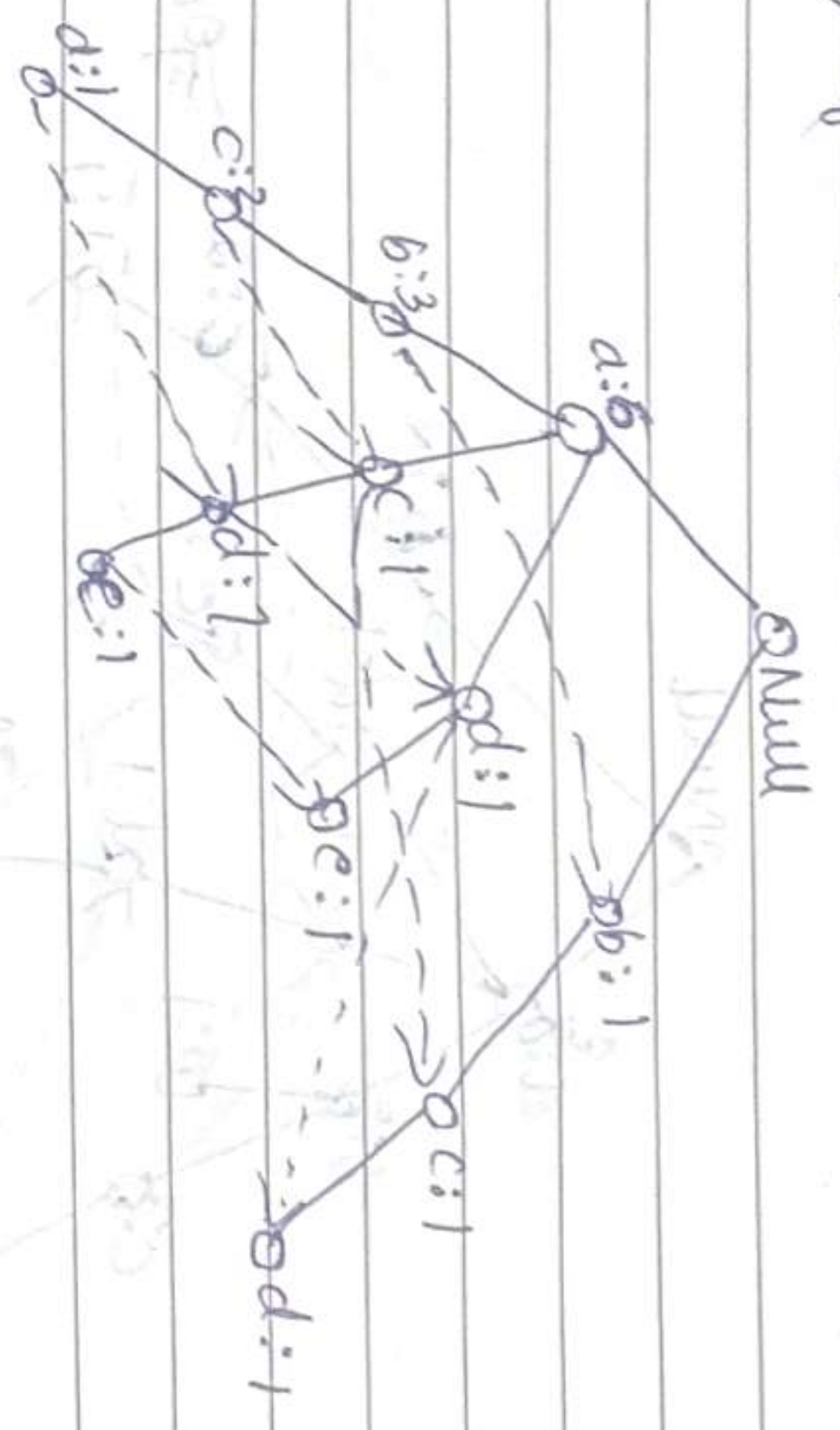
v) after TID = 5



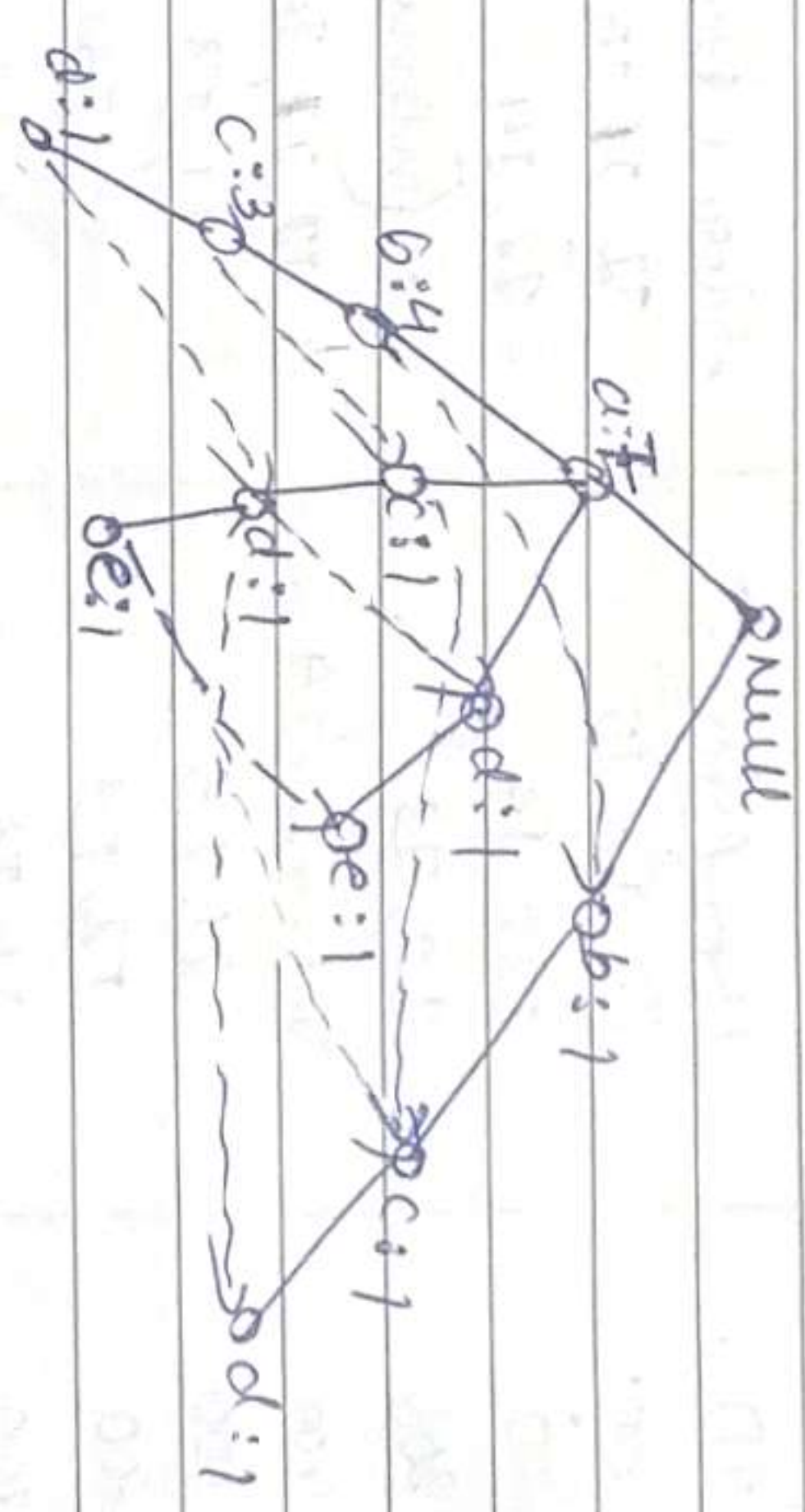
vi) after TID = 6.



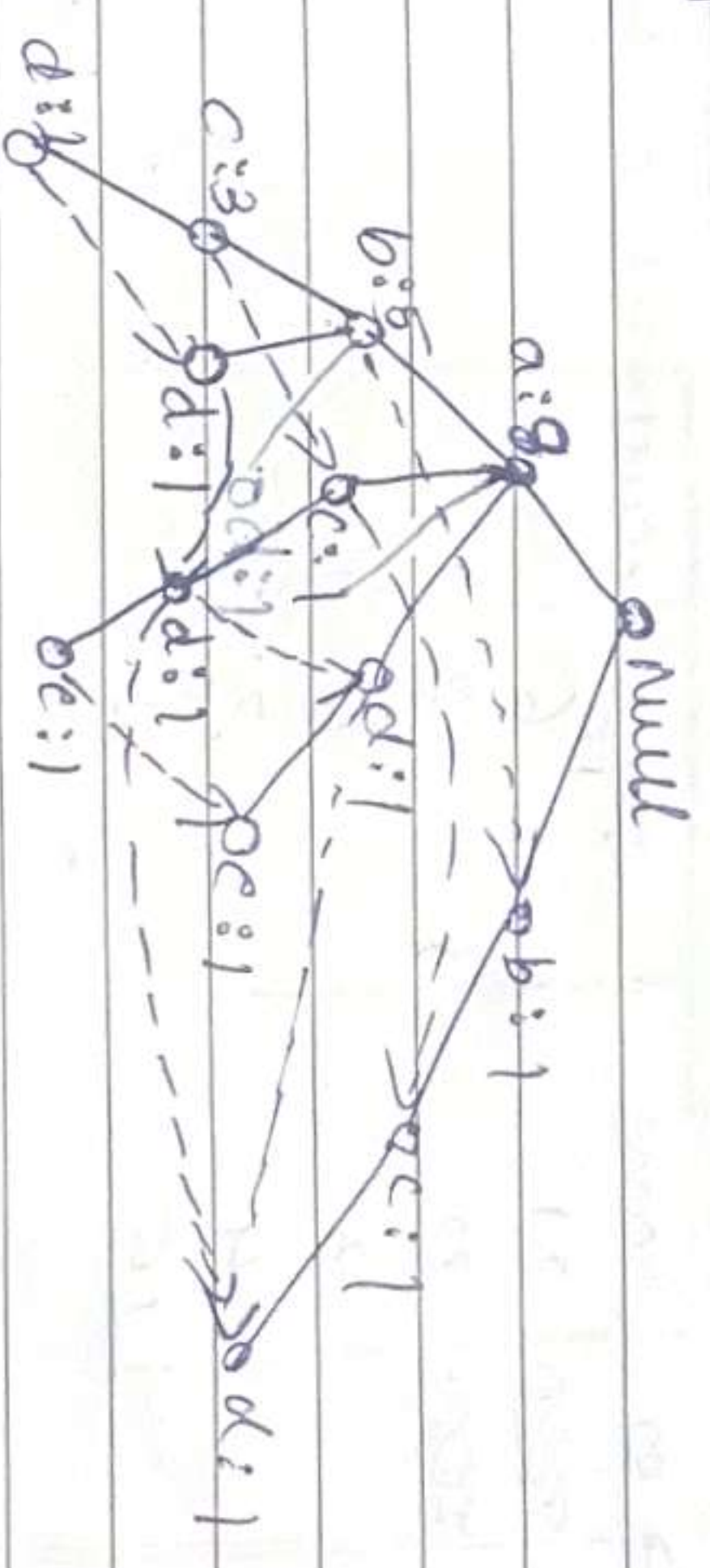
vii) after TID = 7.



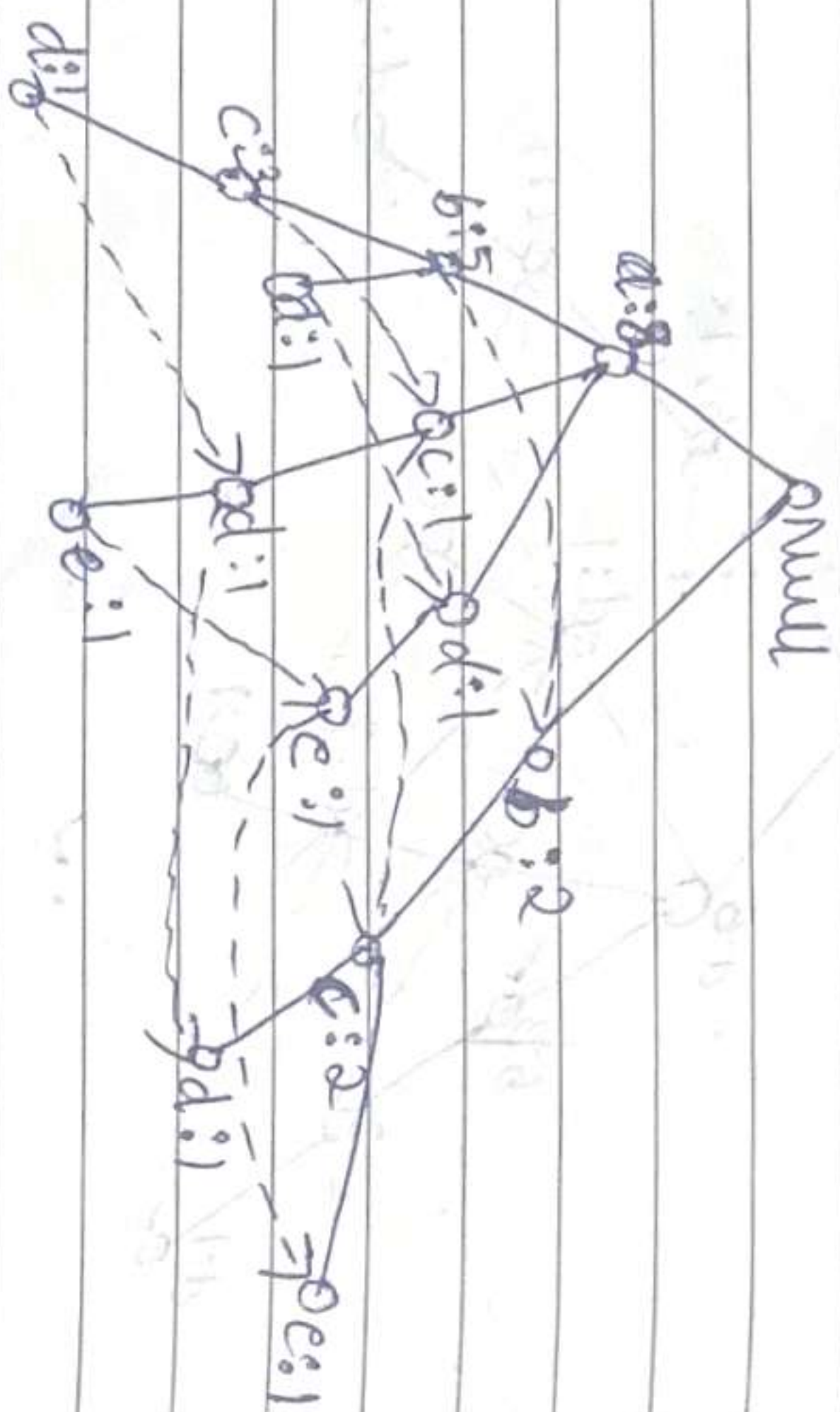
viii) after TID = 8.



ix) after TID = 9.



8) After TID = 10



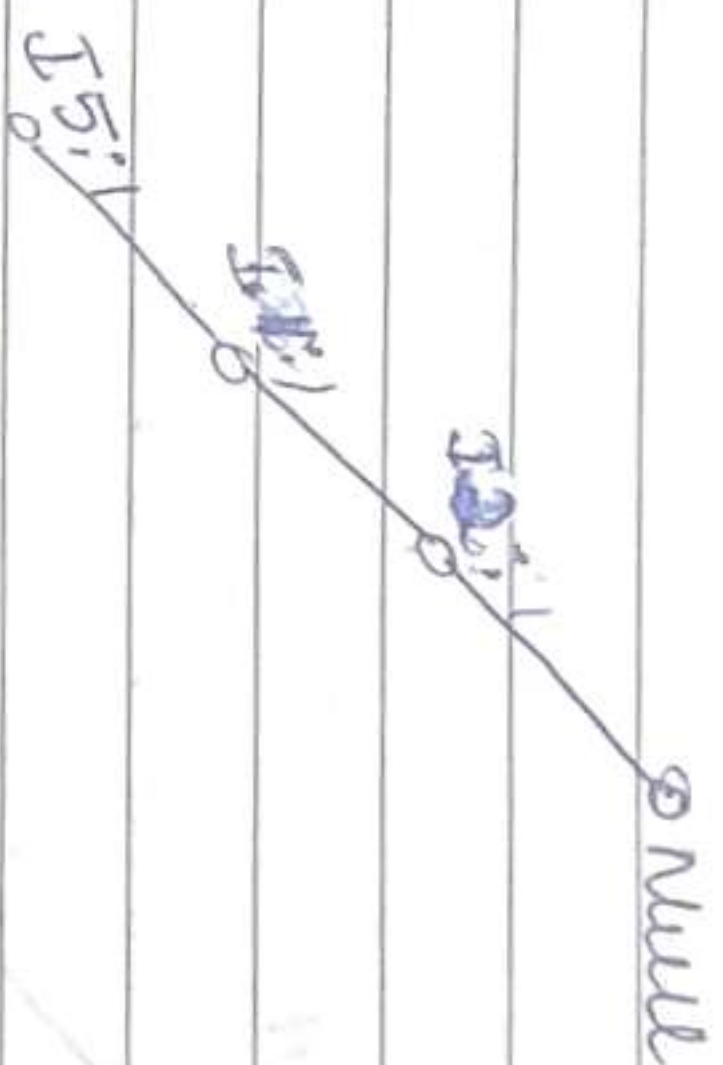
TID	List of Items	Ordered Items
T100	I4, I2, I5	I2, I1, I5
T200	I2, I4	I2, I4
T300	I2, I3	I2, I3 (Interchange 22)
T400	I1, I2, I4	I2, I1, I4
T500	I1, I3	I1, I3
T600	I2, I3	I2, I3
T700	I1, I3	I1, I3
T800	I1, I2, I3, I5	I2, I1, I3, I5
T900	I1, I2, I3	I2, I1, I3

Items	Support count
I1	6
I2	7
I3	6
I4	2
I5	2

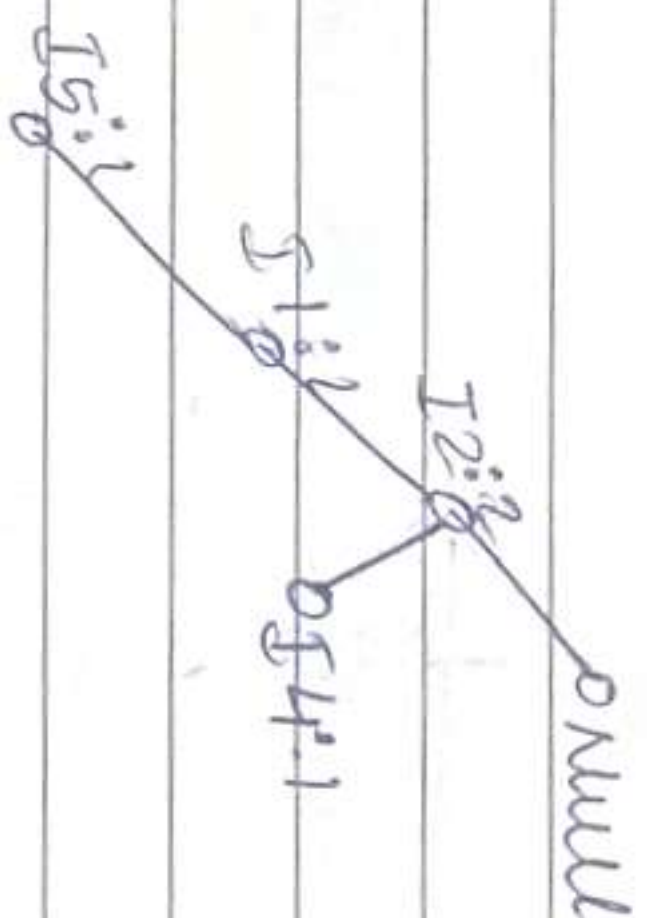
Friday
Descending order

Items	Support count
I2	7
I1	6
I3	6
I4	2
I5	2

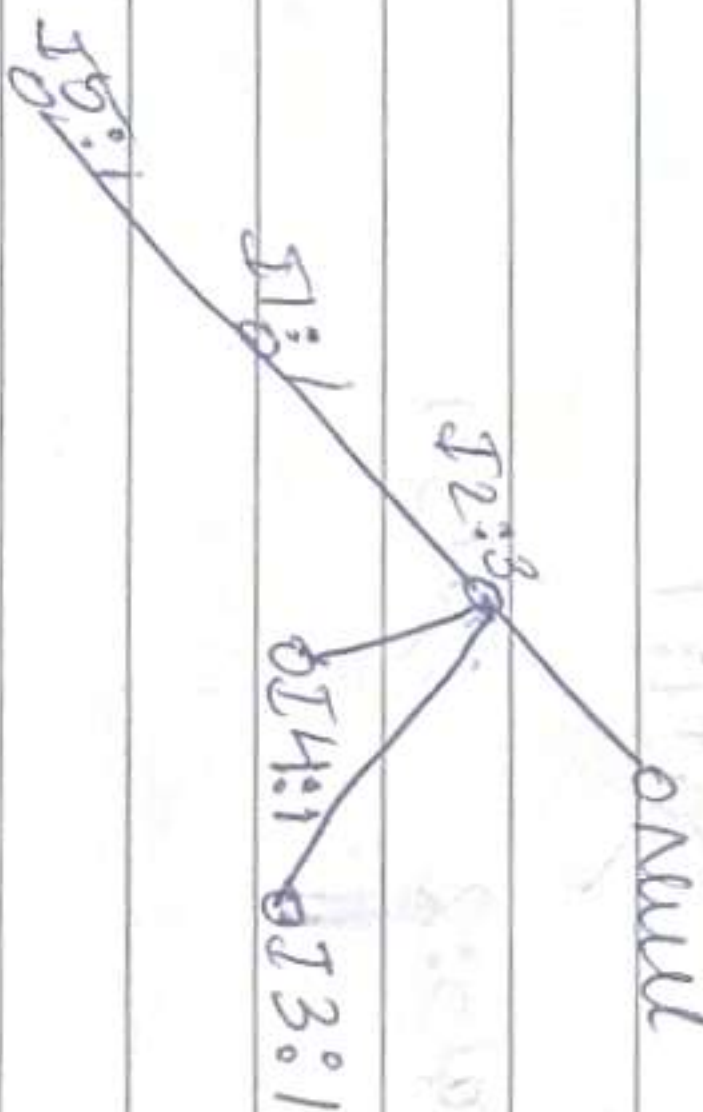
9) After TID = 100



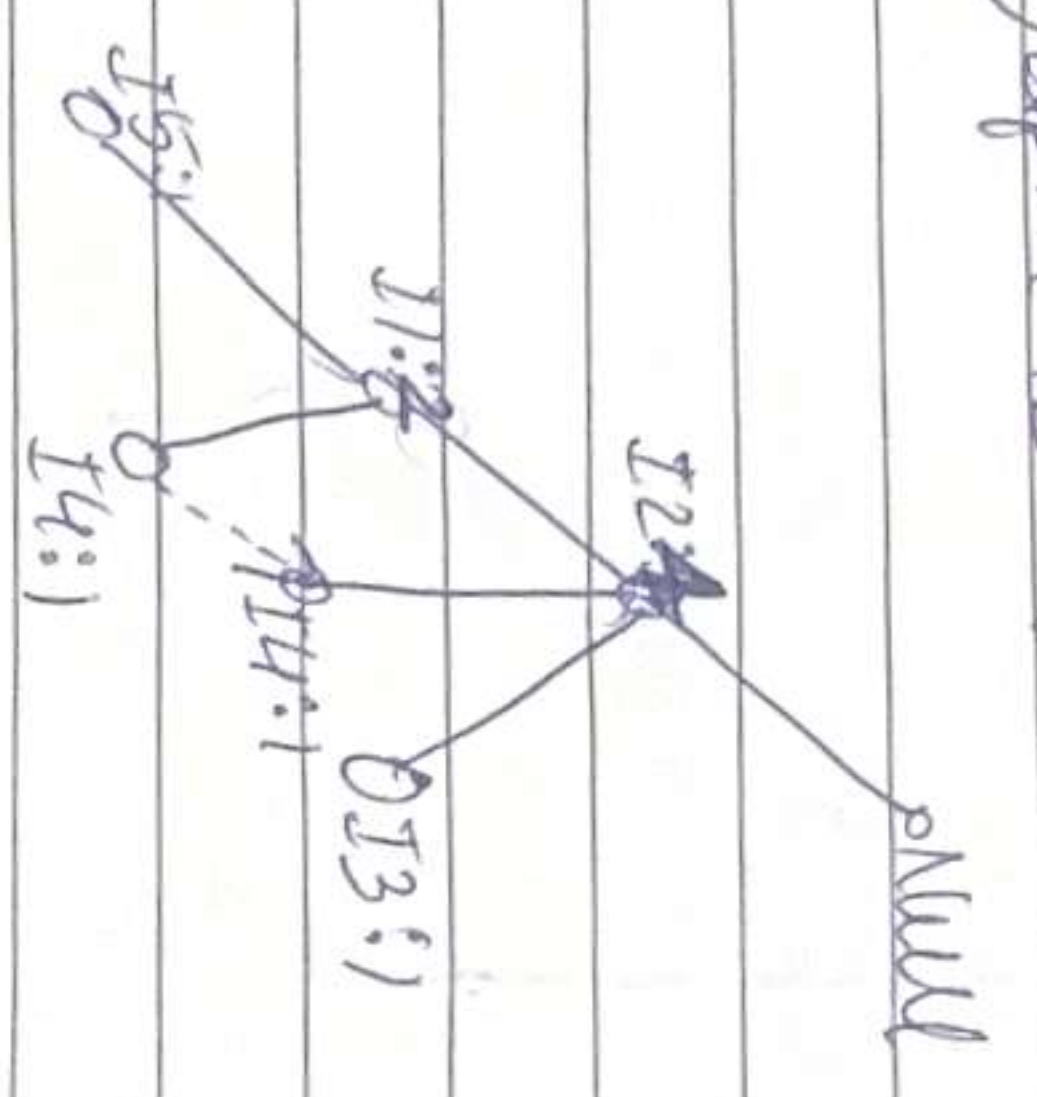
10) After TID = 1200



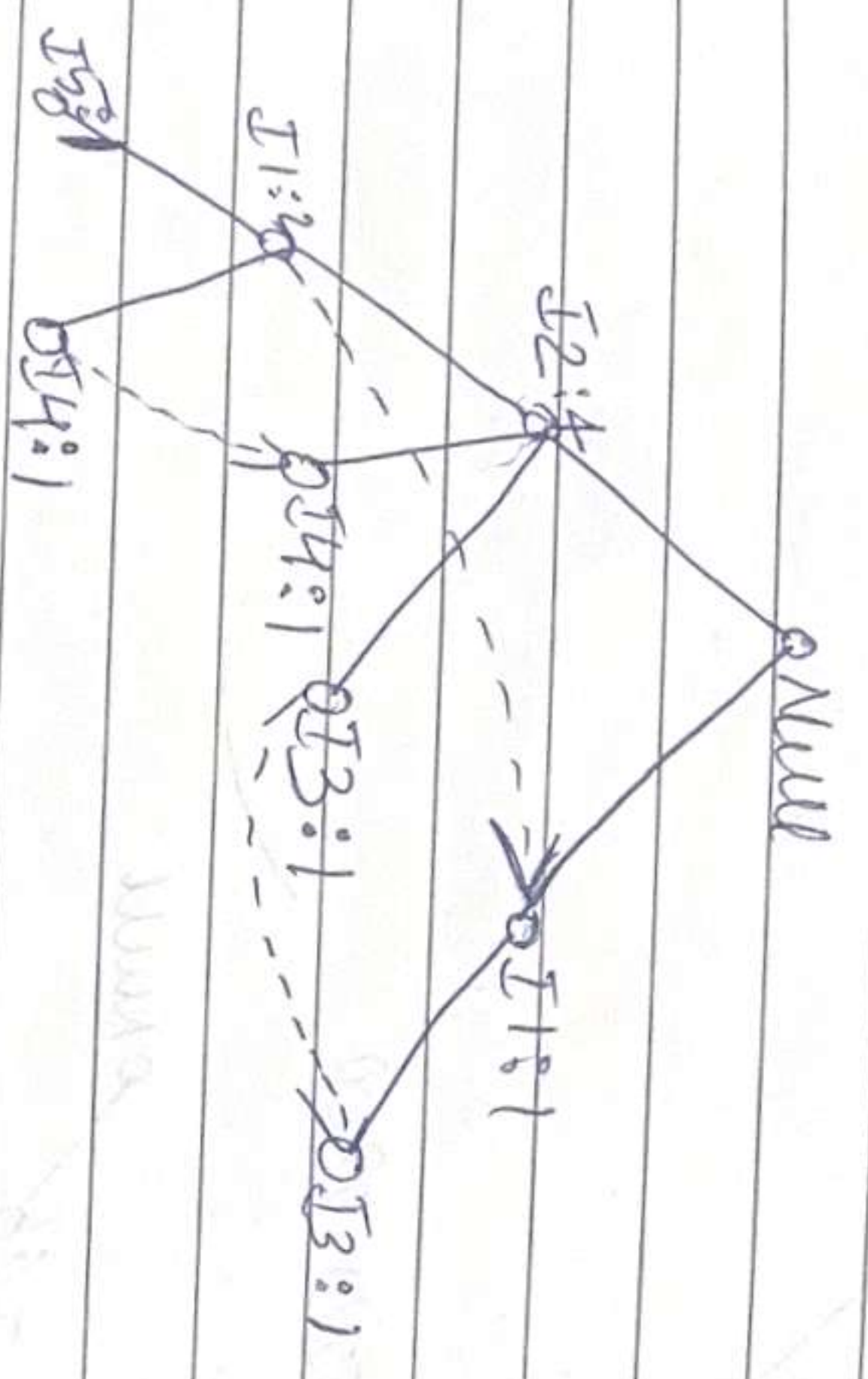
11) After TID = 1300



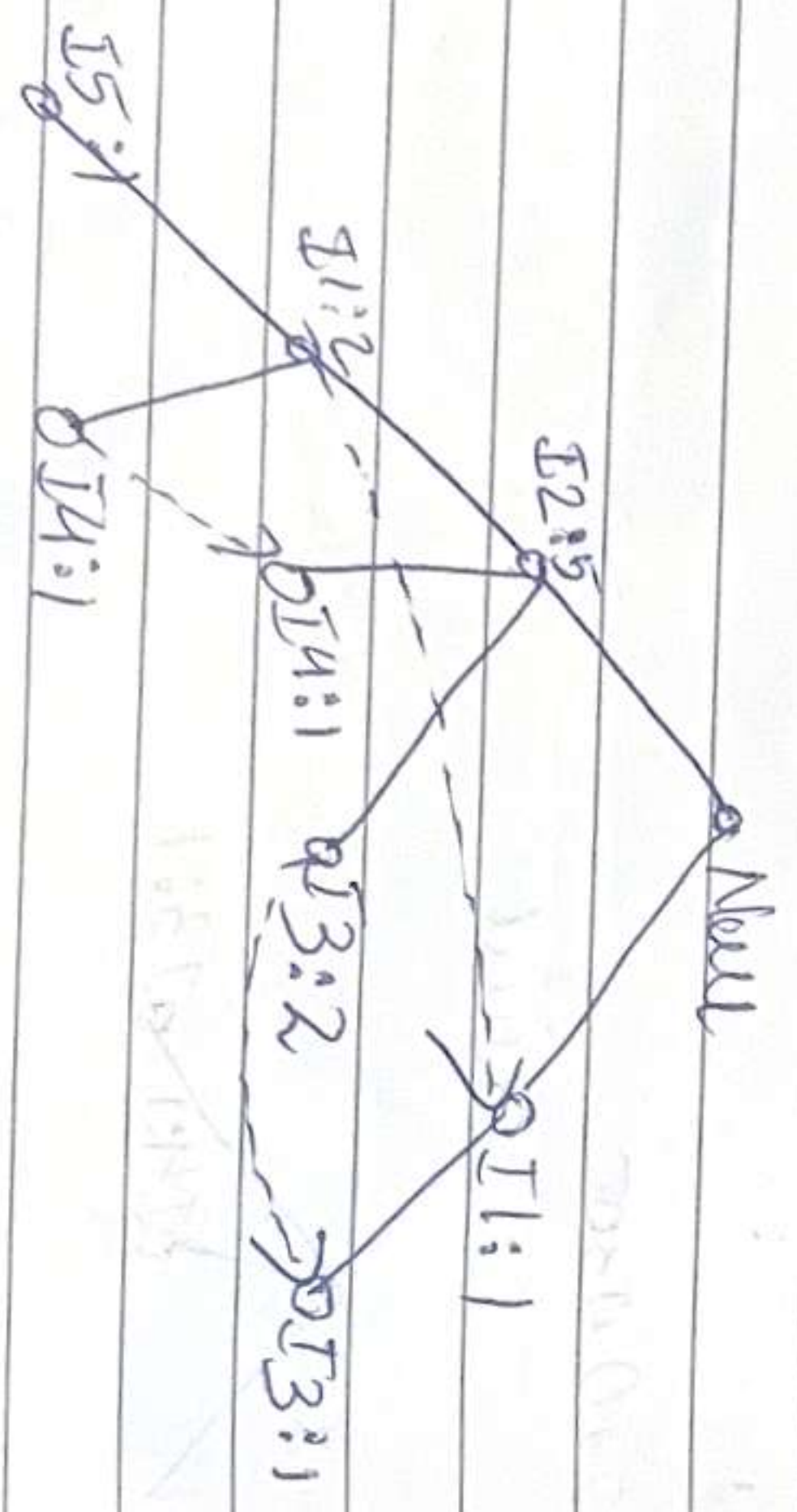
9V) After TID, T400



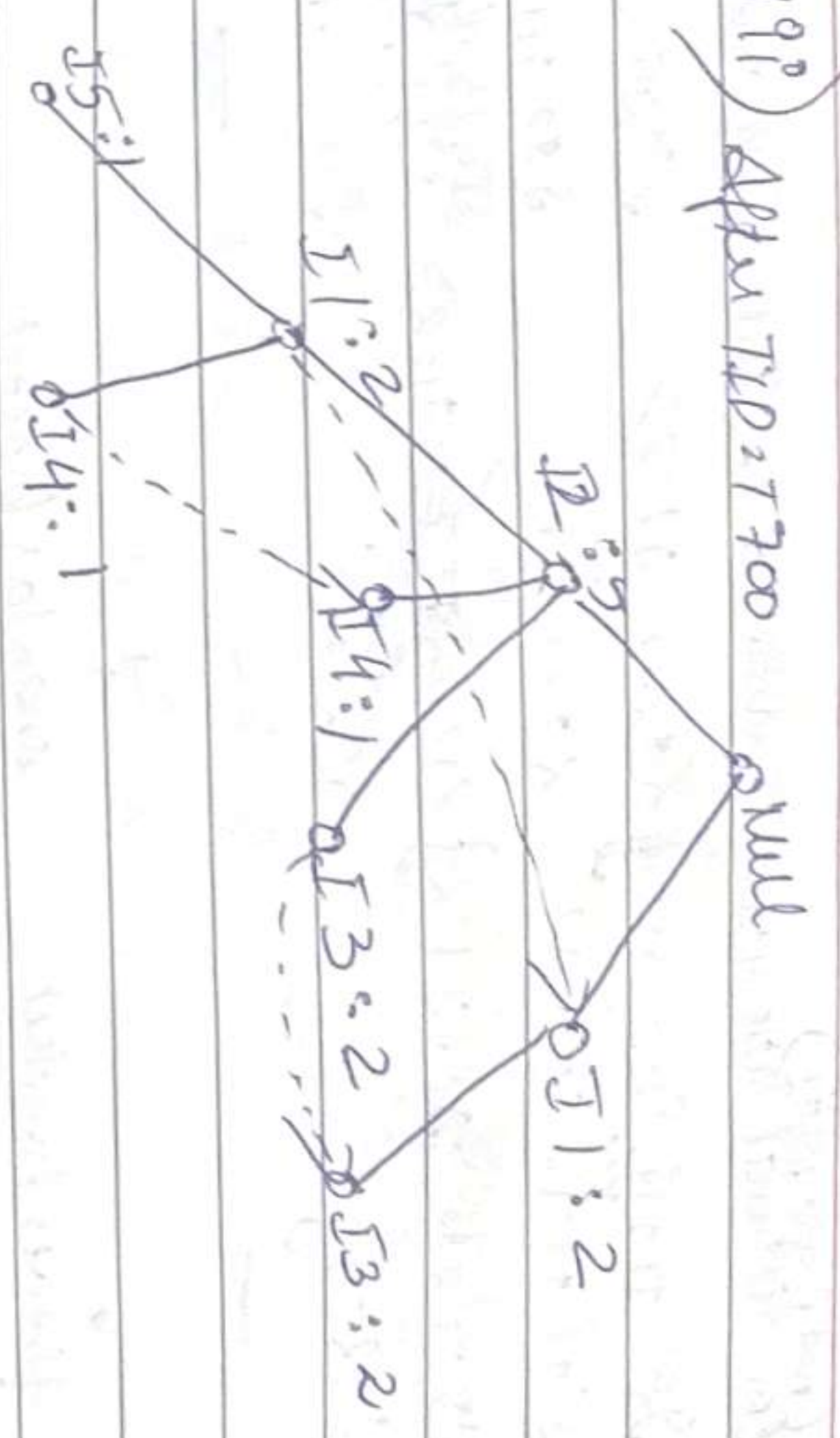
1) After TID, T500



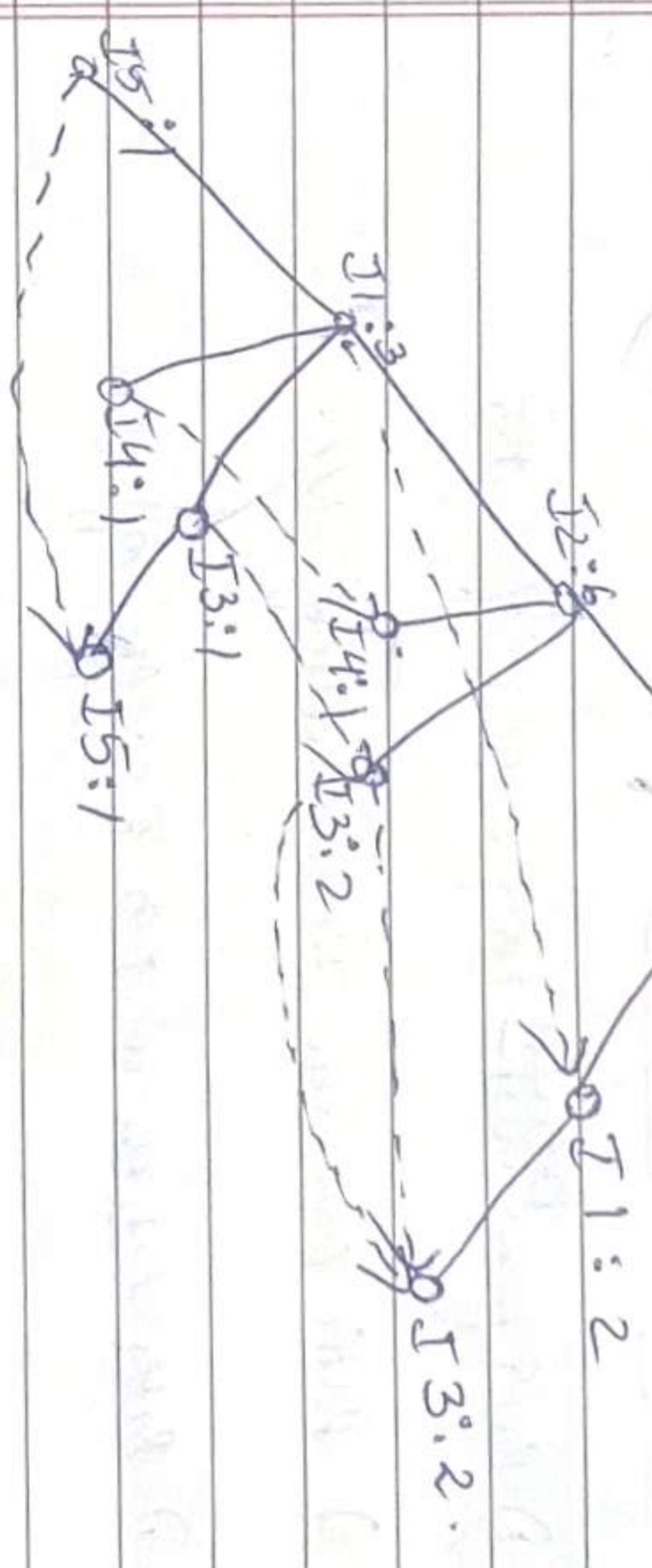
VI) After TID, T600



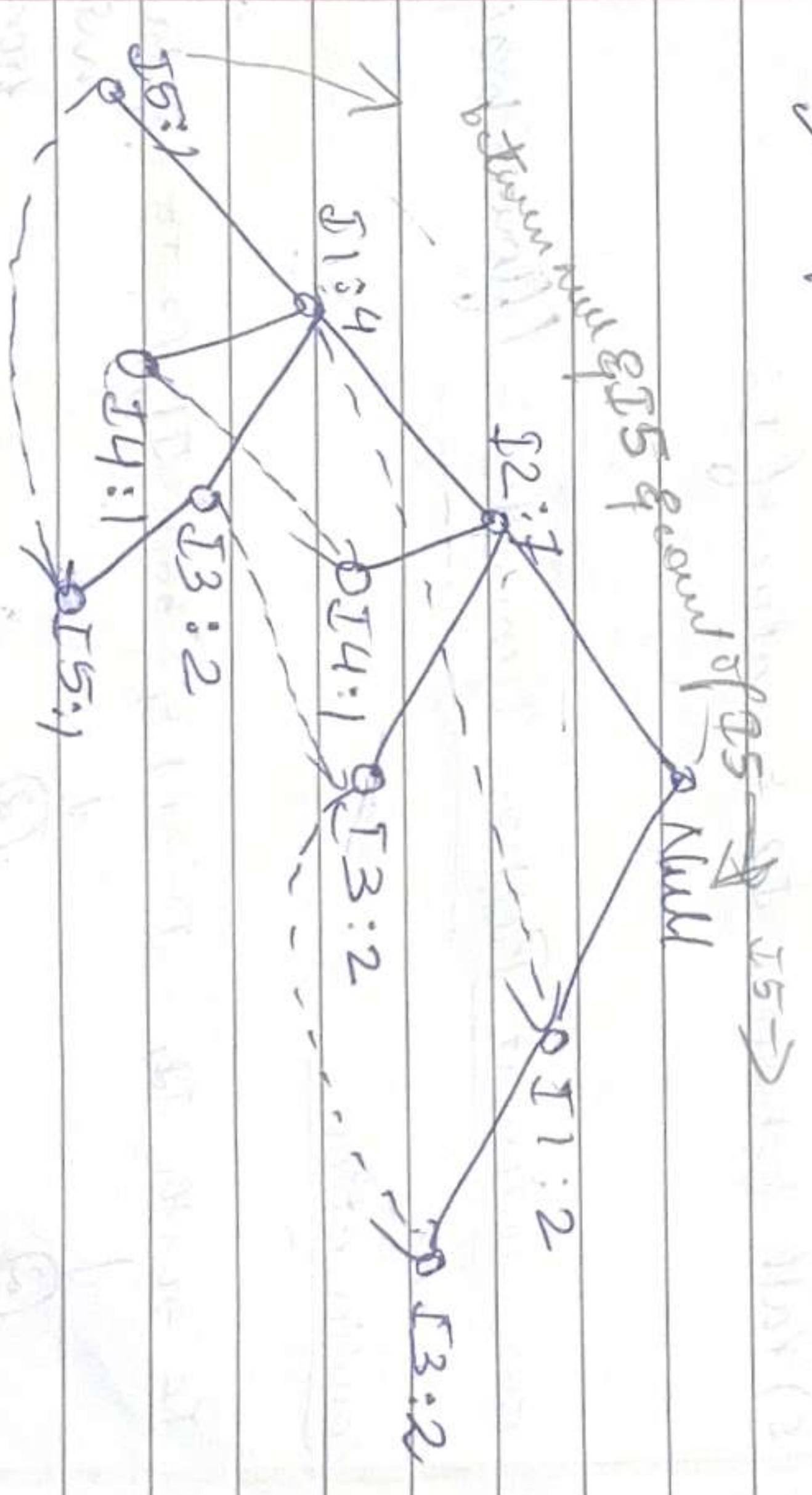
VII) After TID, T700



VIII) After TID, T800



IX) After TID, T900



Let's see always (-)

Item	Conditional Path	Conditional Path	Frequent Path
I5	$\{I2, I1:1, I3:1\}$	$\{I2:2, I1:2\}$	$\{I2, I5:2, I1, I3:2\}$
I4	$\{I2, I1:1, I3:1\}$	$\{I2:2, I1:2\}$	$\{I2, I4:2, I1, I3:2\}$
I3	$\{I2, I1:1, I3:1\}$	$\{I2:2, I1:2\}$	$\{I2, I3:2, I1, I4:2\}$
I1	$\{I2:4, I3:4\}$	$\{I2:4, I1:4\}$	$\{I2, I1:4, I3:4\}$
I2	$\{I2:4, I3:4\}$	$\{I2:4, I1:4\}$	$\{I2, I2:4, I1:4\}$

fewer bracket angular bracket

Conditional Path (Frequent Path)

- 1) Path between I5 & value of I5.
- 2) Path between I4 & value of I4.
- 3) Path between I3 & value of I3.
- 4) Path between I1 & value of I1.
- 5) Path between I2 & value of I2.

Frequent Path (Frequent Path)

- 1) I5 with I3, I5 with I1 & I2, I5 with I4
- 2) I4 with I1 & I2, I4 with I3
- 3) I3 with I1 & I2, I3 with I4
- 4) I1 with I2 & I3, I1 with I4
- 5) I2 with I1 & I3, I2 with I4

2) I4 $\rightarrow$ I2, I4 $\rightarrow$ I3, I4 $\rightarrow$ I1, I4 $\rightarrow$ I5

3) I2, I3 $\rightarrow$ 4, I1, I3 $\rightarrow$ 4, I1, I2, I3 $\rightarrow$ 4

Conditional Path (Frequent Path) if it is minimum
equal. (Angular bracket) $\rightarrow$ false check if it satisfies
Conclusion
Frequent 3 items are I1, I2, I3 and I1, I2, I5

4) Frequent Itemset for the following
transactional dataset using Apriori
principle & FP growth algorithm.

TID	Itemset
100	$\{M, O, N, K, E, Y\}$
200	$\{D, O, N, K, E, Y\}$
300	$\{M, A, K, E\}$
400	$\{M, O, C, K, Y\}$
500	$\{C, O, K, I, E\}$

Apriori Algorithm